

**IMPLEMENTATION AND EVALUATION OF A HYBRID
CLASSICAL-QUANTUM BLOCKCHAIN USING GHZ-BASED
 Θ -PROTOCOL VALIDATION**

PHASE I REPORT

Submitted by

ALFRED SAM D	230701026
BRIJITH MANIKANDAN P	230701058
KASILINGAM M	230701145

*in partial fulfillment for the award of the degree
of*

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**



**RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
ANNA UNIVERSITY: CHENNAI 600 025**

OCTOBER 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled "**IMPLEMENTATION AND EVALUATION OF A HYBRID CLASSICAL-QUANTUM BLOCKCHAIN USING GHZ-BASED @ -PROTOCOL VALIDATION**" is the bonafide work of **ALFRED SAM D (230701026)**, **BRIJITH MANIKANDAN P (230701058)** and **KASILINGAM M (230701145)**, a who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

SIGNATURE

Dr. Manoranjini J, Ph.D.

Professor and Head,
Department of Computer Science and
Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

SIGNATURE

Dr.V.Karthick

Associate Professor,
Department of Computer Science and
Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner

External Examiner

ABSTRACT

Blockchain systems require a reliable consensus mechanism to establish agreement among distributed participants before transactions are committed to a shared ledger. Conventional blockchain systems provide decentralized transaction management and cryptographic protection, but the development of quantum computing creates long-term concerns regarding traditional cryptographic security. Existing quantum blockchain approaches have largely focused on theoretical concepts involving quantum entanglement and quantum states, with limited implementation of complete software pipelines that combine transaction processing, validator management, quantum verification, and ledger synchronization.

The proposed **Hybrid Classical-Quantum Blockchain System** combines conventional blockchain operations with a quantum validation layer to provide an executable software-based framework. The system follows a modular three-layer architecture consisting of the **User/Blockchain Application Layer, Quantum Interface Layer, and Qiskit Execution Layer**. The blockchain layer manages transaction submission, SHA-256 hashing, mempool processing, validator selection, block creation, and ledger management, while the quantum layer performs the required quantum validation using a six-validator network.

The system uses six validator nodes, with each validator associated with a qubit in a shared six-qubit GHZ state. The quantum validation process establishes correlations among the participating validators and performs measurement operations using Qiskit and Qiskit Aer. The resulting binary measurement outcomes are collectively evaluated through an XOR-based parity mechanism. The measured parity is compared with the expected protocol condition, and only a transaction that satisfies the required validation condition is permitted to proceed toward block commitment.

A controlled mempool mechanism is used to manage candidate transactions before validation, while deterministic verifier selection provides a consistent validation workflow. After successful quantum validation, the transaction is incorporated into a classical blockchain block containing transaction information, cryptographic hashes, validator details, and consensus information. The generated block is subsequently synchronized across the six participating validators, maintaining a common representation of the shared ledger. The quantum validation mechanism was evaluated through twenty independent trials in an ideal, noise-free simulation environment. The experiments achieved complete agreement with the expected validation condition, demonstrating consistent operation of the implemented quantum validation process. The average execution time was approximately **3.25 seconds per trial**. These results demonstrate the functional feasibility of integrating a simulated quantum validation mechanism with conventional blockchain processing.

The implemented system provides a foundation for hybrid classical-quantum distributed ledger research by combining transaction management, cryptographic hashing, validator coordination, quantum validation, block creation, and ledger synchronization within a single modular platform. However, the current implementation is based on quantum simulation and therefore does not represent physical quantum-hardware performance. Future development can address hardware noise, scalability, and stronger integration between classical transaction data and quantum validation information.

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E., F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. MANORANJINI J, Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work.

We convey our sincere and deepest gratitude to our internal guide, **Dr.V.Karthick** Associate Professor, Department of Computer Science and Engineering for her valuable guidance, constructive advice, and constant encouragement throughout the course of the project. We thank our Project Coordinator, Assistant Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

Finally, we extend our heartfelt gratitude to our parents, family members, and friends for their constant moral support, patience, and motivation which inspired us to achieve this milestone.

ALFRED SAM D :	230701026
BRIJITH MANIKANDAN P :	230701058
KASILINGAM M :	230701145

Chapter / Section	Title	Page No.
	ABSTRACT	2
	ACKNOWLEDGEMENT	3
	LIST OF TABLES	8
	LIST OF FIGURES	9
CHAPTER 1	INTRODUCTION	10
1.1	General	10
1.2	Objective	11
1.3	Existing System	11
1.4	Proposed System	12
CHAPTER 2	LITERATURE SURVEY	14
2.1	Literature Review	14
2.2	Comparison and Discussion	15
2.3	Research Gap	16
2.4	Conclusion	17
CHAPTER 3	SYSTEM DESIGN	18
3.1	System Architecture	18
3.2	System Requirements	19
3.2.1	Software Requirements	19
3.2.2	Hardware Requirements	20

3.2.3	Quantum Simulation Requirements	21
3.2.4	Blockchain Network Requirements	22
3.3	Blockchain Network Design	22
3.4	Quantum Validation Architecture	23
3.5	System Workflow	24
3.5.1	Transaction Processing Workflow	24
3.5.2	Quantum Validation Workflow	25
3.5.3	Block Creation and Ledger Replication	25
3.6	UML and System Diagrams	25
CHAPTER 4	PROJECT DESCRIPTION	26
4.1	Methodology	26
4.2	System Modules	26
4.2.1	User Transaction Module	27
4.2.2	Classical Blockchain Module	28
4.2.3	Validator Network Module	28
4.2.4	Quantum Validation Module	28
4.2.5	GHZ State Generation Module	28
4.2.6	θ -Protocol Validation Module	29
4.2.7	Consensus Decision Module	29
4.2.8	Shared Ledger Replication Module	30
4.3	Classical Blockchain Layer	30
4.3.1	Transaction and Mempool Processing	30
4.3.2	SHA-256 Hashing	30
4.3.3	Block Formation	30

4.3.4	Block Chaining	31
4.4	Quantum Validation Layer	31
4.4.1	Qubit and Quantum Circuit Model	31
4.4.2	Six-Qubit GHZ State	32
4.4.3	Measurement Angle Generation	32
4.4.4	Quantum Measurement	32
4.4.5	XOR Parity Calculation	32
4.5	θ -Protocol Consensus Mechanism	32
4.6	Mathematical Formulation	33
4.7	Algorithm and Pseudocode	34
CHAPTER 5	IMPLEMENTATION AND RESULT DISCUSSION	36
5.1	Implementation Environment	36
5.2	System Implementation	36
5.3	Transaction Processing Results	37
5.4	Quantum Validation Results	37
5.5	Blockchain and Ledger Results	38
5.6	Performance Analysis	39
5.7	Shared Ledger Implementation	40
5.8	Website Implementation and Results	40
5.9	Experiments and Results	43
5.10	Performance Analysis	44
5.11	Result Discussion	45
5.12	Advantages of Implemented System	46
5.13	Limitations	46

CHAPTER 6	CONCLUSION AND FUTURE WORK	48
6.1	Conclusion	48
6.2	Future Work	49
	REFERENCES	50
	APPENDIX	52

LIST OF TABLES

Table No.	Title	Page No.
Table 2.1	Comparison of Existing Approach And Proposed Approach	16
Table 3.1	Software Requirements	20
Table 3.2	Hardware Requirements	20
Table 4.1	System Modules and Functions	27
Table 5.1	Implementation Environment	37
Table 5.2	Sample Quantum Validation Result	39
Table 5.3	Quantum Validation Experimental Results	44
Table 5.4	Performance Metrics	45
Table A.1	Key Implementation Parameters	56

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 3.1	System Architecture Diagram	18
Figure 3.2	Validation Network Diagram	21
Figure 3.3	Blockchain Network	23
Figure 4.1	Quantum Circuit Diagram	31
Figure 5.1	Website Screenshot – Blockchain and Validator Information	41
Figure 5.2	Website Screenshot – Transaction and Block Details	42
Figure 5.3	Website Screenshot – Quantum Validation and Consensus Details	43

CHAPTER I

INTRODUCTION

1.1 GENERAL

Blockchain technology is a distributed ledger technology that enables transactions to be recorded and maintained across a network of participating nodes without depending entirely on a centralized authority. In a conventional blockchain, transactions are submitted to the network, validated according to predefined rules, grouped into blocks, and linked with previously created blocks using cryptographic hashing. The distributed nature of the ledger allows multiple participating nodes to maintain synchronized copies of the blockchain, thereby providing transparency, consistency, and resistance to unauthorized modification of recorded information.

The development of quantum computing has introduced new research possibilities as well as challenges for existing computational and cryptographic systems. Quantum computing utilizes principles of quantum mechanics, including superposition, entanglement, and quantum measurement, to process information in ways that differ from conventional classical computing. Although practical large-scale quantum computing is still developing, the potential impact of quantum technologies on existing cryptographic and distributed systems has encouraged research into quantum-assisted and quantum-resistant blockchain architectures.

The proposed work focuses on the implementation and evaluation of a **Hybrid Classical–Quantum Blockchain Using GHZ-Based 0-Protocol Validation**. The system combines conventional blockchain operations with a quantum-based validation mechanism. The classical layer is responsible for transaction processing, SHA-256 hashing, block formation, block chaining, validator participation, and shared ledger replication. The quantum layer introduces a six-qubit GHZ-state-based validation process in which measurement outcomes from the participating quantum subsystems are used to evaluate a predefined parity condition.

The proposed architecture consists of a user node and a network of six consensus validators. When a transaction is submitted, it passes through the classical transaction-processing stage before being subjected to the defined quantum validation procedure. A six-qubit GHZ state is generated, and measurement operations are performed using protocol-defined measurement angles. The resulting binary measurement values are combined using an XOR operation to obtain the measured parity. This value is then compared with the expected parity to determine whether the quantum validation condition has been satisfied.

The system is implemented as a software-based prototype and the quantum validation process is evaluated using a quantum simulation environment. Therefore, the proposed work demonstrates the integration of quantum validation concepts with classical blockchain mechanisms rather than claiming deployment on a physical quantum computer. The resulting architecture provides a

controlled environment for studying the interaction between blockchain transaction processing, distributed validators, quantum-state-based validation, and replicated ledger management.

1.2 OBJECTIVE

The primary objective of the proposed project is to implement and evaluate a hybrid classical–quantum blockchain architecture in which a quantum validation mechanism is integrated with conventional blockchain operations. The system is designed to preserve the fundamental operations of a classical blockchain while introducing a quantum validation stage based on a six-qubit GHZ state and the θ -protocol.

The proposed system aims to provide a complete transaction-processing workflow beginning with user transaction submission and continuing through classical transaction processing, cryptographic hashing, validator participation, quantum validation, block creation, and ledger replication. The classical blockchain component is responsible for maintaining the transaction and block structure, while the quantum component is used to evaluate the predefined validation condition through quantum measurement outcomes.

Another important objective is to model a distributed validation network containing six consensus validators and associate the validators with the six quantum subsystems used in the validation process. The six-qubit GHZ state is prepared as part of the quantum validation stage, after which measurement operations are performed using the required measurement angles. The resulting binary outputs are processed using XOR parity, and the measured parity is compared with the expected parity to determine whether the validation condition has been satisfied.

The project also aims to experimentally evaluate the correctness and execution performance of the quantum validation mechanism in a simulated environment. Through this implementation, the project demonstrates how quantum information concepts can be incorporated into a classical blockchain architecture while maintaining the conventional blockchain mechanisms required for transaction processing and ledger management.

1.3 EXISTING SYSTEM

Conventional blockchain systems rely primarily on classical computing mechanisms for transaction processing, cryptographic operations, consensus, and ledger management. In a typical blockchain architecture, a transaction is generated by a user and submitted to the network. The transaction is then processed and validated according to the rules defined by the blockchain protocol. Valid transactions are subsequently grouped into blocks, while cryptographic hashing is used to maintain the relationship between consecutive blocks.

The use of cryptographic hash functions is an important component of conventional blockchain systems. Hashing generates a fixed-length representation of transaction or block information, allowing changes to the underlying data to be detected. In addition to hashing, blockchain networks employ consensus mechanisms through which participating nodes agree on the validity and ordering

of transactions. Once consensus is reached, the validated block can be added to the blockchain and replicated across participating nodes.

Although conventional blockchain systems provide distributed transaction validation and ledger management, their consensus and validation processes are based entirely on classical computational mechanisms. Quantum information principles such as quantum entanglement and quantum measurement are not inherently incorporated into the conventional blockchain validation process. Consequently, traditional blockchain architectures do not provide a framework for experimentally studying quantum-state-based consensus conditions together with classical transaction and ledger operations.

The existing architecture therefore provides a strong foundation for distributed data management but does not combine classical blockchain processing with an additional quantum validation layer. The proposed project addresses this area by retaining the classical blockchain workflow while introducing a quantum validation mechanism as an additional stage in the transaction-validation process.

1.4 PROPOSED SYSTEM

The proposed system introduces a hybrid architecture that combines a conventional classical blockchain layer with a quantum validation layer. Instead of replacing the existing blockchain mechanisms, the proposed approach integrates the quantum component with the existing transaction and block-processing workflow. This allows classical blockchain operations such as transaction processing, SHA-256 hashing, block generation, block chaining, and ledger replication to continue while the quantum mechanism performs the defined validation operation.

In the proposed architecture, a user first submits a transaction to the blockchain system. The transaction is received by the classical processing layer and prepared for validation. The required transaction information is processed using the classical blockchain mechanism, including SHA-256 hashing. The transaction then proceeds through the validator network, which consists of six consensus validators participating in the validation process.

The quantum validation stage uses a six-qubit GHZ state corresponding to the six-validator configuration. The GHZ state provides the required quantum correlation between the participating qubits. Measurement settings are generated according to the θ -protocol, and each qubit is measured using its corresponding measurement angle. The measurements produce six binary outputs representing the results obtained from the participating quantum subsystems.

The six measurement outcomes are then combined using an XOR operation to calculate the measured parity. If the measurement outcomes are represented as:

$$\mathbf{Y} = [y_1, y_2, y_3, y_4, y_5, y_6]$$

the measured parity is:

$$\mathbf{P}_{\text{measured}} = y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6$$

Here, $\oplus$ represents the XOR (exclusive OR) operation.

The measured parity is compared with the expected parity defined by the validation protocol. When the measured parity satisfies the required condition, the quantum validation stage is considered successful. The transaction can then proceed to the classical block-creation stage, after which the validated block is replicated across the participating validator ledgers.

The proposed architecture therefore establishes an interaction between classical blockchain processing and quantum validation. The classical layer manages the transaction and blockchain lifecycle, while the quantum layer provides an experimental validation mechanism based on the GHZ-state correlation and parity condition. This separation also makes the architecture modular, allowing the classical and quantum components to be studied independently as well as as an integrated system.

The proposed system is intended primarily as a software-based experimental implementation. The quantum circuits are executed using a quantum simulation environment rather than physical quantum hardware. Therefore, the results demonstrate the behavior and validation capability of the implemented protocol under the simulated environment. The quantum mechanism should be understood as a validation component for the defined quantum correlation and parity condition rather than as a replacement for classical cryptographic authentication of all blockchain data.

The major distinction between the existing and proposed systems is therefore the integration of a quantum validation stage into the conventional blockchain transaction lifecycle. The proposed architecture maintains the fundamental properties of classical blockchain processing while providing a framework for investigating the use of quantum states and measurements in distributed validation.

CHAPTER II

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The integration of blockchain technology with quantum computing has emerged as an important area of research due to the increasing interest in secure distributed systems and the future capabilities of quantum computation. Conventional blockchain systems depend on cryptographic techniques, distributed consensus, and replicated ledgers to provide integrity and trust among participating nodes. At the same time, quantum computing introduces new computational principles such as superposition, entanglement, and quantum measurement, creating opportunities to investigate new approaches for blockchain validation and consensus.

Existing research on blockchain and quantum computing can broadly be considered from two perspectives. The first perspective focuses on the **security of blockchain systems against future quantum attacks**, particularly the possibility that sufficiently powerful quantum computers could affect cryptographic algorithms used by existing blockchain networks. The second perspective investigates how **quantum computing itself can be incorporated into blockchain architectures**, including quantum consensus, quantum communication, quantum cryptography, and quantum-enhanced validation mechanisms.

Research related to quantum-resistant blockchain architectures has primarily concentrated on replacing or strengthening classical cryptographic mechanisms with algorithms that are expected to remain secure in the presence of quantum computers. These approaches are important for the long-term security of blockchain networks, but they generally continue to use classical computing for transaction processing and consensus. Therefore, quantum computing is considered mainly as a potential threat that the blockchain must resist rather than as a computational component that can participate in the validation process.

Another area of research investigates quantum cryptographic techniques for secure communication between distributed participants. Quantum key distribution and related approaches use properties of quantum mechanics to establish or protect cryptographic keys. Such approaches demonstrate the usefulness of quantum information in secure distributed environments. However, quantum communication-based security and quantum blockchain validation represent different objectives, since the proposed project focuses on incorporating a quantum validation mechanism into the blockchain transaction lifecycle.

Quantum entanglement has also attracted research interest for distributed quantum systems. Entangled quantum states allow correlations to be established between spatially separated quantum subsystems. In the proposed work, this concept is investigated through a six-qubit GHZ state, where the participating quantum subsystems are correlated and their measurement results are used to evaluate a predefined parity condition.

The use of GHZ states provides a suitable foundation for investigating multi-party quantum correlations. A GHZ state can be represented generally as a superposition of the all-zero and all-one computational states. For a six-qubit system, the state can be represented as

$$|\text{GHZ}_6\rangle = 1/\sqrt{2} (|000000\rangle + |111111\rangle)$$

The proposed system utilizes this multi-qubit correlated state as part of its quantum validation mechanism. Instead of using the quantum state merely for communication or key generation, the project investigates whether the measurement outcomes can be used to evaluate a predefined consensus condition among multiple validator participants.

The literature also demonstrates the increasing use of quantum simulation frameworks for developing and evaluating quantum algorithms. Quantum simulators provide an environment in which quantum circuits can be constructed and executed without requiring access to physical quantum processors. This is particularly useful for experimental blockchain research because the quantum component can be integrated with conventional software systems and evaluated under controlled conditions.

The proposed project builds upon these research directions by combining classical blockchain processing with a quantum validation mechanism. The work does not attempt to replace classical blockchain cryptography or claim that the quantum component independently provides complete blockchain security. Instead, it investigates the integration of a GHZ-based θ -protocol validation process with transaction processing, validator participation, block generation, and shared ledger replication.

2.2 COMPARISON AND DISCUSSION

The existing research demonstrates that blockchain and quantum computing can be connected in several different ways. Quantum-resistant blockchain research primarily addresses the threat posed by quantum computers to existing cryptographic mechanisms. Quantum communication research focuses on improving the security of communication between distributed participants, while quantum blockchain research investigates the use of quantum principles in blockchain-related operations.

The proposed project differs from these approaches by concentrating specifically on the integration of a **quantum validation mechanism with a classical blockchain transaction lifecycle**. The system maintains the conventional blockchain functions while introducing a quantum validation stage based on a six-qubit GHZ state and a θ -protocol parity condition.

TABLE 2.1 – COMPARISON OF EXISTING RESEARCH APPROACHES AND PROPOSED SYSTEM

Research Direction	Primary Focus	Quantum Component	Blockchain Integration
Conventional Blockchain	Distributed ledger and consensus	Not used	Full classical implementation
Quantum-Resistant Blockchain	Protection against quantum attacks	Post-quantum cryptography	Classical blockchain
Quantum Cryptographic Systems	Secure communication/key management	Quantum communication	Partial/related
Quantum Blockchain Research	Quantum-based blockchain concepts	Quantum computation	Varies
Proposed System	Hybrid transaction validation	Six-qubit GHZ state and θ -protocol	Classical blockchain + quantum validation

The comparison indicates that the proposed system is primarily concerned with the **experimental integration of quantum validation into an existing classical blockchain workflow**. The classical component remains responsible for transaction and ledger management, while the quantum component evaluates the defined parity-based validation condition.

An important aspect of the proposed approach is the use of multiple validators represented through multiple qubits. The six-validator configuration provides a direct relationship between the distributed validation network and the six-qubit quantum state. The measurement outcomes obtained from these qubits are combined through XOR parity, providing a clearly defined mechanism for determining whether the quantum validation condition has been satisfied.

The architecture also provides a separation between blockchain data management and quantum validation. This separation is useful because the quantum mechanism does not need to replace the existing blockchain data structures. Instead, it functions as an additional validation stage before the validated transaction is incorporated into the shared ledger.

2.3 RESEARCH GAP

From the reviewed research directions, it can be observed that blockchain and quantum computing have been investigated extensively as individual technologies and as interconnected research areas. However, there remains scope for practical software-based architectures that demonstrate how a

quantum validation mechanism can be integrated directly into the transaction lifecycle of a classical blockchain.

Many quantum-related blockchain approaches concentrate on quantum-resistant cryptography, quantum communication, or theoretical quantum blockchain models. These approaches address important aspects of future blockchain security and quantum technology, but they do not necessarily provide a complete implementation that connects user transactions, validator participation, quantum measurement, consensus evaluation, block creation, and replicated ledger management within a single experimental workflow.

The proposed project addresses this gap by implementing a hybrid architecture in which classical blockchain processing and quantum validation operate as interconnected components. The six-validator network is associated with a six-qubit GHZ state, and the measurement results are evaluated using an XOR parity condition. This provides a concrete software-level mechanism through which the quantum validation process can be observed and evaluated.

Another aspect of the research gap is the practical evaluation of such a hybrid architecture in a simulated environment. The proposed system provides an experimental framework in which quantum validation can be executed using a quantum simulator while the remaining blockchain operations are performed using conventional software components. This makes it possible to evaluate the behavior of the proposed mechanism without requiring physical quantum hardware.

The project therefore contributes an implementation-oriented approach for studying the interaction between distributed blockchain validation and quantum-state-based consensus conditions.

2.4 SUMMARY

The literature survey demonstrates that blockchain and quantum computing represent complementary areas of research with significant potential for future distributed systems. Existing research has investigated quantum-resistant cryptography, quantum communication, quantum security, and quantum blockchain concepts. These studies provide the foundation for investigating how quantum principles can be incorporated into blockchain architectures.

Based on the reviewed approaches, the proposed project focuses on a specific integration of classical blockchain processing and quantum validation. The system retains conventional transaction processing, cryptographic hashing, block creation, and shared ledger management while introducing a six-qubit GHZ-state-based θ -protocol for quantum validation.

The literature review therefore establishes the motivation for the proposed hybrid architecture and identifies the need for practical experimental implementations that connect quantum validation with the complete blockchain transaction lifecycle. The following chapter presents the system design, architecture, requirements, validator network, workflow, and diagrams used to implement the proposed system.

CHAPTER III

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system is designed as a hybrid architecture that combines a conventional classical blockchain network with a quantum validation mechanism. The architecture is organized into multiple functional components that work together to process transactions, perform validation, create blocks, and maintain a synchronized shared ledger. The system consists of a user node, a validator network containing six consensus validators, a classical blockchain processing layer, a quantum validation layer, and a shared ledger.

The user node acts as the starting point of the transaction lifecycle. A transaction submitted by the user is received by the blockchain application and forwarded to the processing and validation components. The classical blockchain layer manages the conventional operations associated with the transaction, including transaction handling, SHA-256 hashing, block formation, block chaining, and ledger management. The validator network contains six participating validator nodes. These validators are responsible for participating in the transaction-validation process. The six-validator configuration also corresponds to the six quantum subsystems used by the quantum validation layer. This provides a relationship between the distributed validator network and the quantum circuit used for validation.

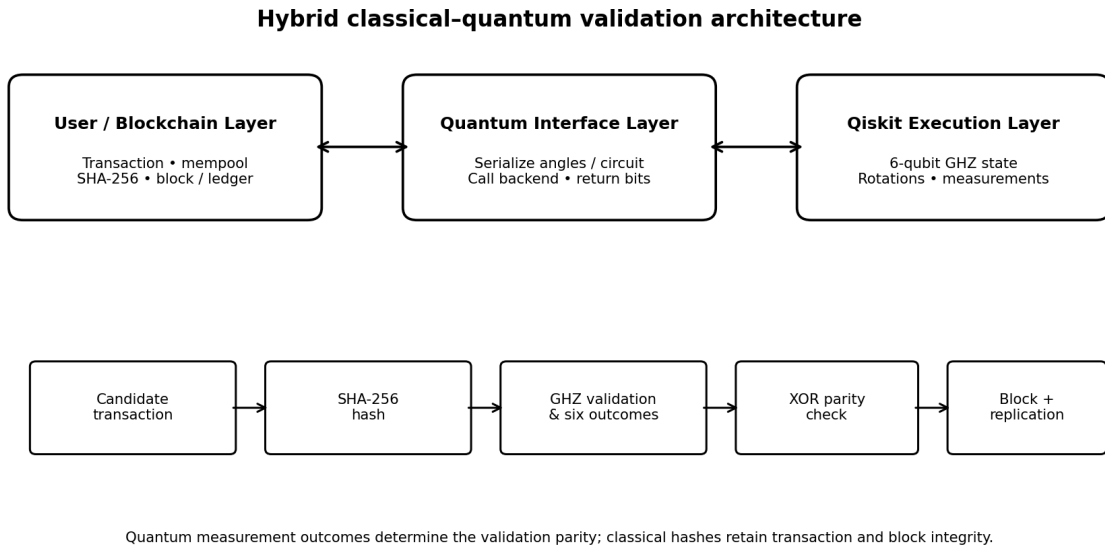


FIGURE 3.1 – SYSTEM ARCHITECTURE DIAGRAM

The quantum validation layer is responsible for executing the GHZ-based θ -protocol. A six-qubit GHZ state is prepared, and measurement operations are performed using the measurement settings

generated according to the protocol. The resulting measurement outcomes are represented as binary values and combined using XOR parity. The resulting parity is compared with the expected parity to determine whether the quantum validation condition has been satisfied.

When the required validation condition is satisfied, the transaction proceeds through the classical blockchain process for block creation. The newly generated block is then added to the blockchain and replicated across the participating validator ledgers. This creates a shared ledger representation in which the validator nodes maintain synchronized blockchain information.

The architecture therefore separates the system into classical and quantum processing components while allowing them to interact during the validation stage. The classical component handles blockchain data and ledger operations, whereas the quantum component provides the experimental validation mechanism.

3.2 SYSTEM REQUIREMENTS

The proposed system requires both classical computing resources and quantum simulation software to implement and evaluate the hybrid blockchain architecture. Since the quantum validation mechanism is implemented using simulation rather than physical quantum hardware, the required environment consists primarily of conventional computing hardware, blockchain software components, programming libraries, and a quantum computing framework.

3.2.1 SOFTWARE REQUIREMENTS

The software environment consists of the programming and development tools required to implement the classical blockchain components and the quantum validation mechanism. A suitable programming environment is required for implementing transaction processing, blockchain data structures, hashing operations, validator management, quantum circuit construction, and result processing.

The quantum component requires a quantum computing framework capable of constructing and executing quantum circuits through simulation. The implementation uses Qiskit-based quantum circuit operations and a simulator to execute the quantum validation process. The simulator enables the six-qubit circuit to be evaluated without requiring access to physical quantum hardware.

The web-based component of the system provides an interface through which users can interact with the blockchain application and observe the transaction and validation results.

TABLE 3.1 – SOFTWARE REQUIREMENTS

Category	Requirement	Purpose
Programming Environment	Python	Core implementation
Quantum Framework	Qiskit	Quantum circuit construction and execution
Quantum Simulator	Qiskit-compatible simulator	Simulated quantum execution
Hashing	SHA-256	Cryptographic hashing
Web Interface	Project web application	User interaction
Development Environment	Code editor / IDE	Development and testing

3.2.2 HARDWARE REQUIREMENTS

The proposed system does not require a physical quantum computer because the quantum validation mechanism is evaluated using a simulator. The primary hardware requirement is therefore a conventional computer capable of running the blockchain application, web interface, quantum circuit simulation, and supporting development tools.

The required computing resources depend on the size of the simulated quantum circuit and the number of executions performed during experimentation. Since the proposed architecture uses six qubits, the quantum simulation requirement remains suitable for execution on a conventional development system.

TABLE 3.2 – HARDWARE REQUIREMENTS

Component	Minimum / Project Configuration	Purpose
Processor	INTEL i5	Application and simulation execution
RAM	16 GB	Runtime memory
Storage	512 GB	Software and project files
Operating System	WINDOWS	Development environment

Network	Required for web/network components	Communication
Quantum Hardware	Not required	Simulation-based execution

3.2.3 QUANTUM SIMULATION REQUIREMENTS

The quantum component of the proposed system is implemented using a simulated quantum environment. The simulator provides a classical computational representation of the quantum circuit and allows the proposed GHZ-state-based validation mechanism to be executed and evaluated.

The quantum simulation environment is responsible for preparing the required qubits, constructing the six-qubit GHZ state, applying the measurement operations, and obtaining the binary measurement results. These results are subsequently processed by the classical application to calculate the XOR parity and determine the validation outcome.

The use of simulation is particularly suitable for this project because the objective is to demonstrate and evaluate the proposed validation protocol without depending on the availability of physical quantum hardware. The simulation environment also provides a controlled setting for repeating experiments and measuring execution behavior.

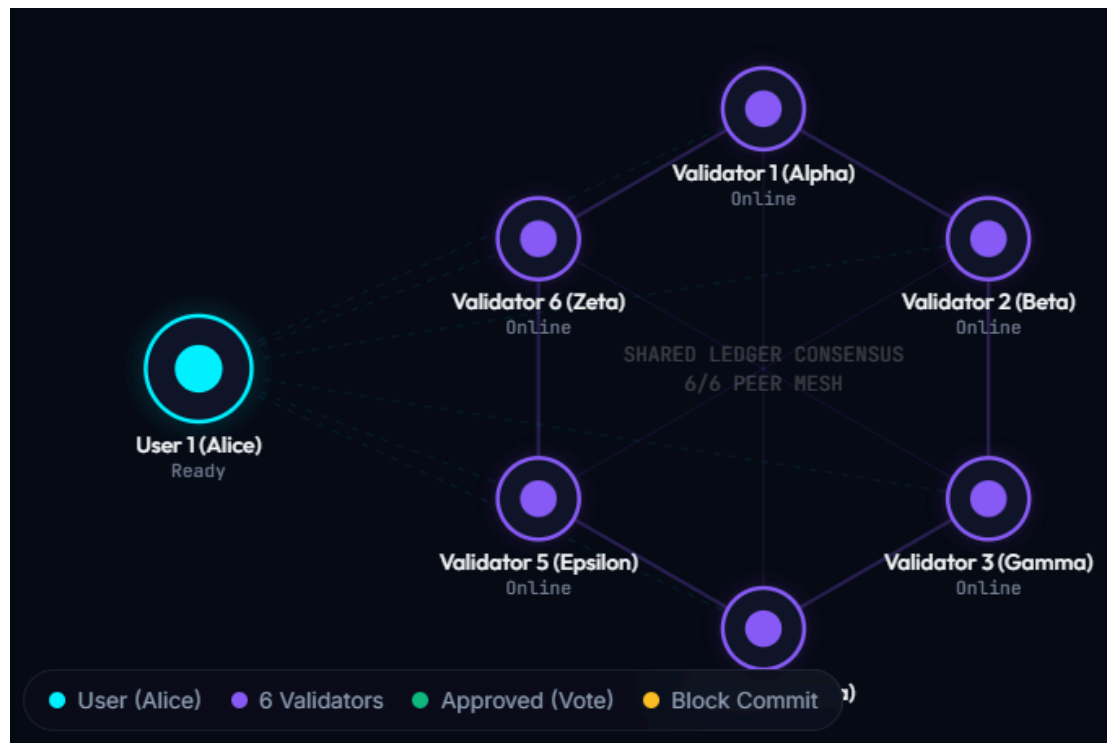


FIGURE 3.2 – VALIDATION NETWORK DIAGRAM

The validation network diagram represents the relationship between the six classical validators and the corresponding six quantum subsystems. This diagram should be placed here because it directly explains how the validator network participates in the quantum validation process.

3.2.4 BLOCKCHAIN NETWORK REQUIREMENTS

The blockchain component requires a mechanism for managing transactions, validating blocks, maintaining block relationships, and replicating the resulting ledger across the validator network. Each block contains the information required to represent the validated transaction and maintain its relationship with the preceding block.

The system uses cryptographic hashing to maintain the integrity of the blockchain structure. SHA-256 is used to generate a hash representation of the relevant blockchain information. The generated hash contributes to the identification and chaining of blocks within the classical blockchain layer.

The validator network provides the distributed component of the architecture. The six validators participate in the validation process, and successful transactions can result in a block being created and reflected across the shared ledger maintained by the validator network.

3.3 BLOCKCHAIN NETWORK DESIGN

The blockchain network is designed around a user node and six validator nodes. The user node is responsible for initiating transactions, while the validator nodes participate in the validation and ledger-maintenance process. The network is designed to demonstrate how a transaction can move from submission through validation and finally into a replicated shared ledger.

When a user submits a transaction, the transaction is received by the application and prepared for processing. The transaction information is then processed by the classical blockchain component, where the required hash is generated. The transaction is subsequently associated with the validator network for the validation stage.

The six validators provide the distributed participants required by the proposed architecture. Their participation is connected to the six-qubit quantum validation mechanism. The quantum validation results are used to determine whether the transaction satisfies the defined validation condition before the corresponding block is finalized.

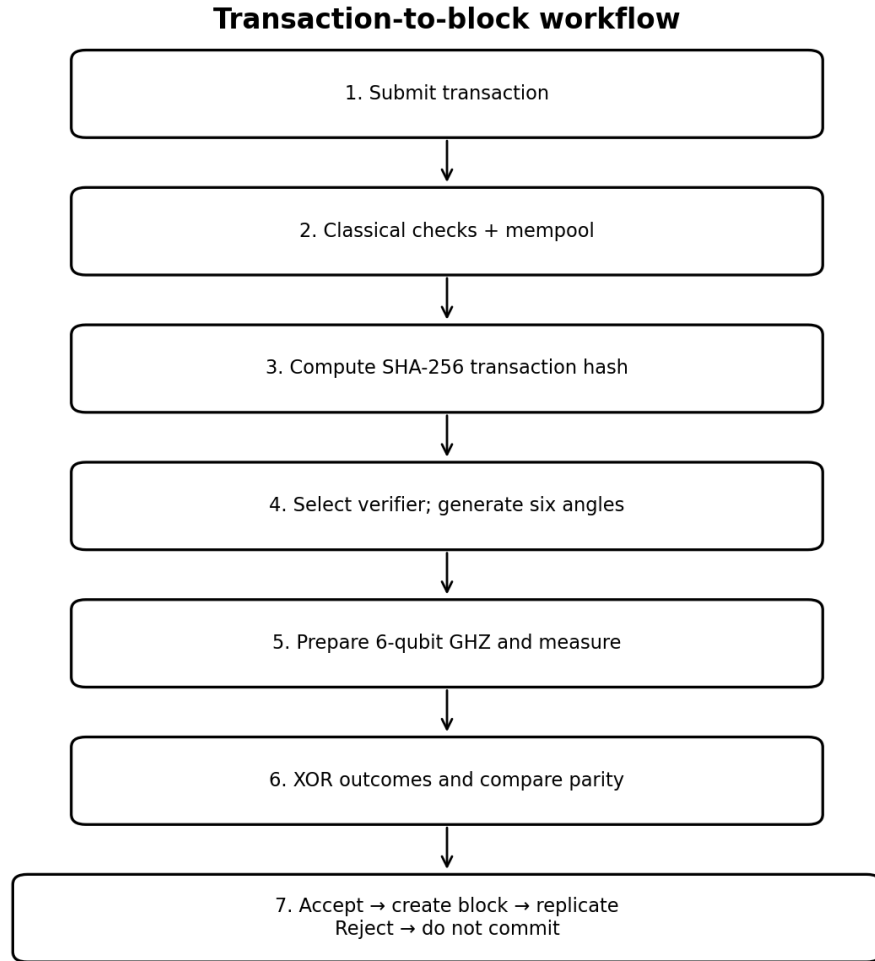


FIGURE 3.3 – BLOCKCHAIN NETWORK

Once validation is successfully completed, the transaction is included in a new block. The block contains the required blockchain information and is connected to the previous block through the hashing mechanism. The resulting blockchain state is then reflected across the validator nodes, producing the shared ledger.

3.4 QUANTUM VALIDATION ARCHITECTURE

The quantum validation architecture forms the main distinguishing component of the proposed hybrid blockchain. It introduces a quantum circuit into the classical blockchain transaction-validation lifecycle. The quantum circuit consists of six qubits corresponding to the six participating validator positions.

The validation process begins with the preparation of a six-qubit GHZ state. The GHZ state establishes quantum correlations between the participating qubits. Once the state has been prepared,

measurement operations are performed according to the measurement settings generated by the θ -protocol.

The measurement results are represented using binary values. If the six results are represented by $y_1, y_2, y_3, y_4, y_5, y_6$, they are combined through an XOR operation to obtain the measured parity. The parity result provides the value used by the validation mechanism to determine whether the quantum condition has been satisfied.

The quantum validation architecture therefore consists of state preparation, measurement-angle assignment, quantum measurement, measurement-result collection, parity calculation, and validation decision. The final validation result is then passed back to the classical blockchain workflow.

3.5 SYSTEM WORKFLOW

The complete system workflow begins when a user submits a transaction through the application interface. The transaction is received by the blockchain processing layer and prepared for validation. The classical component performs the required transaction processing and hashing operations before the transaction enters the validator-based validation stage.

The six validators participate in the validation process. At the quantum stage, a six-qubit GHZ state is prepared and measurement operations are performed using the protocol-defined measurement settings. The six quantum measurement outcomes are collected and processed to calculate the XOR parity.

The calculated parity is compared with the expected parity. When the required condition is satisfied, the transaction is considered successfully validated by the proposed quantum validation mechanism. The transaction can then proceed to the block-generation stage.

Following successful block generation, the new block is connected to the existing blockchain through the hashing mechanism. The resulting blockchain state is then replicated across the validator nodes to maintain the shared ledger.

3.5.1 TRANSACTION PROCESSING WORKFLOW

The transaction-processing workflow begins at the user interface, where transaction information is entered and submitted. The application receives the transaction and forwards it to the blockchain processing component. The transaction is then prepared for validation and associated with the appropriate blockchain information.

The transaction undergoes the required classical processing before being passed to the validator network. The blockchain component generates the required cryptographic hash and maintains the information necessary for subsequent block creation.

The processed transaction is then subjected to the proposed quantum validation mechanism. The result of the quantum validation determines whether the transaction can proceed toward block formation.

3.5.2 QUANTUM VALIDATION WORKFLOW

The quantum validation workflow begins with the preparation of the six-qubit GHZ state. The six qubits represent the six validator positions in the proposed network. After state preparation, the measurement settings are generated according to the θ -protocol.

Each qubit is measured using its corresponding measurement setting, producing a binary output. The six outputs are collected into a measurement vector represented as

$$\mathbf{Y} = [y_1, y_2, y_3, y_4, y_5, y_6]$$

The XOR operation is then applied to these values to obtain the measured parity:

$$\mathbf{P}_{\text{measured}} = y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6$$

The calculated parity is compared against the expected parity. If the values satisfy the required protocol condition, the quantum validation is successful. Otherwise, the validation condition is not satisfied.

3.5.3 BLOCK CREATION AND LEDGER REPLICATION

After successful validation, the transaction is processed by the classical blockchain component for block creation. The block contains the required transaction and blockchain information and is associated with the cryptographic hash generated during the classical processing stage.

The newly created block is linked with the previous block through the blockchain's hash relationship. Once the block has been finalized, the resulting blockchain information is replicated across the participating validator ledgers. This produces a synchronized shared ledger representation across the network.

3.6 UML AND SYSTEM DIAGRAMS

The proposed system can be represented using system and workflow diagrams to provide a clear understanding of the interaction between the user, blockchain components, validator network, quantum validation mechanism, and shared ledger. These diagrams provide a visual representation of the architecture and complement the textual description of the system.

The **System Architecture Diagram** presents the overall relationship between the major components of the proposed hybrid blockchain. The **Validation Network Diagram** illustrates the six-validator configuration and its relationship with the quantum validation process. The **Block Diagram** presents the major processing stages involved in transaction validation and block generation.

CHAPTER IV

PROJECT DESCRIPTION

4.1 METHODOLOGY

The proposed project follows a hybrid methodology in which conventional blockchain operations are combined with a quantum validation mechanism. The methodology is designed to maintain the standard transaction and ledger-processing functions of a blockchain while introducing a quantum validation stage based on a six-qubit GHZ state and the θ -protocol. The complete process begins with transaction submission and proceeds through classical processing, validator participation, quantum validation, block creation, and shared ledger replication.

When a user submits a transaction, the transaction is received by the blockchain application and prepared for processing. The classical blockchain layer manages the transaction information and generates the required cryptographic representation using SHA-256 hashing. The transaction is then forwarded to the validation stage, where the six-validator network participates in the proposed validation process.

The quantum validation stage begins by preparing a six-qubit GHZ state. The six qubits correspond to the six validator positions in the proposed network. Measurement angles are then generated according to the θ -protocol. The quantum circuit applies the corresponding measurement operations to the qubits and produces binary measurement outcomes.

The measurement results are collected and represented as a binary vector. The XOR operation is applied to the individual measurement outcomes to calculate the measured parity. The measured parity is then compared with the expected parity defined by the validation protocol. If the required parity condition is satisfied, the transaction passes the quantum validation stage.

Following successful validation, the transaction proceeds to the classical blockchain block-creation stage. The validated transaction is incorporated into a block, and the block is linked to the existing blockchain through the hashing mechanism. The resulting block is subsequently reflected across the participating validator ledgers, producing the shared ledger state.

4.2 SYSTEM MODULES

The proposed system is organized into a set of interconnected modules that collectively implement the hybrid blockchain architecture. Each module performs a specific function within the overall transaction lifecycle, while the interaction between the modules enables the classical blockchain and quantum validation components to operate as a single system.

The major modules include the user transaction module, classical blockchain module, validator network module, quantum validation module, GHZ-state generation module, θ -protocol validation

module, consensus decision module, and shared ledger replication module. The modules are designed to provide a clear separation between transaction management, blockchain processing, quantum validation, and ledger management.

TABLE 4.1 – SYSTEM MODULES AND THEIR FUNCTIONS

Module	Description
User Transaction Module	Receives and processes transactions submitted by the user
Classical Blockchain Module	Performs blockchain transaction and block operations
Validator Network Module	Represents the six participating consensus validators
Quantum Validation Module	Executes the quantum-based validation process
GHZ State Generation Module	Prepares the six-qubit GHZ state
θ -Protocol Module	Performs validation using protocol-defined measurement settings
Consensus Decision Module	Determines whether the validation condition is satisfied
Shared Ledger Module	Maintains the replicated blockchain state

4.2.1 USER TRANSACTION MODULE

The user transaction module provides the entry point for the blockchain system. Through the web-based interface, the user can submit the required transaction information to the application. The submitted information is received by the backend processing component and converted into the appropriate transaction representation required by the blockchain layer.

The module is responsible for initiating the transaction lifecycle and communicating the transaction to the subsequent processing components. Once the transaction is received, it is prepared for classical blockchain processing and validation.

The user interface also provides a convenient way to observe the outcome of the transaction and validation process. The website screenshots presented in Chapter 5 demonstrate the implemented interface and the resulting system outputs.

4.2.2 CLASSICAL BLOCKCHAIN MODULE

The classical blockchain module is responsible for implementing the conventional blockchain functionality of the proposed system. It manages the transaction data, hashing operations, block formation, block chaining, and interaction with the validator network.

SHA-256 hashing is used to generate a cryptographic hash associated with the relevant blockchain information. The hash provides a fixed-length representation that can be used as part of the blockchain structure and helps maintain the relationship between blocks.

After a transaction has successfully passed the required validation process, the classical blockchain module creates the corresponding block and associates it with the previous block. This allows the blockchain to maintain a sequential chain of validated blocks.

4.2.3 VALIDATOR NETWORK MODULE

The validator network module represents the distributed validation participants in the proposed architecture. The system models six consensus validators that participate in the validation process. These validators provide the distributed structure required by the proposed hybrid blockchain.

The six-validator configuration also corresponds to the six quantum subsystems used in the quantum validation process. This relationship provides a connection between the classical distributed network and the quantum circuit.

4.2.4 QUANTUM VALIDATION MODULE

The quantum validation module provides the main quantum-processing component of the proposed system. It receives the validation information required for the quantum circuit and executes the quantum operations through the simulation environment.

The module prepares the quantum state, applies the required measurement operations, collects the measurement outcomes, and provides the results to the consensus decision component. The quantum validation mechanism is therefore integrated with the classical blockchain workflow rather than operating as an independent quantum experiment.

The purpose of this module is to evaluate the defined quantum correlation and parity condition. It should not be interpreted as independently authenticating every classical field of the blockchain transaction.

4.2.5 GHZ STATE GENERATION MODULE

The GHZ state generation module prepares the six-qubit quantum state required by the validation protocol. A GHZ state establishes quantum correlations among multiple qubits and is therefore suitable for representing the multi-participant quantum validation configuration used in the proposed system.

For the six-qubit implementation, the GHZ state can be represented as

$$|\text{GHZ}_6\rangle = 1/\sqrt{2} (|000000\rangle + |111111\rangle)$$

The state is prepared within the quantum circuit before the measurement operations are applied. The generated quantum state provides the correlated quantum resource required for the subsequent validation process.

4.2.6 θ -PROTOCOL VALIDATION MODULE

The θ -protocol validation module performs the quantum measurement process using the measurement settings defined by the proposed protocol. Measurement angles are generated for the participating qubits while satisfying the required protocol relationship.

Each qubit is measured according to its assigned measurement angle. The measurement operation produces a binary output corresponding to the observed quantum result. The six outputs are subsequently collected for parity evaluation.

The measurement stage therefore converts the quantum state and measurement settings into classical binary results that can be processed by the consensus decision component.

4.2.7 CONSENSUS DECISION MODULE

The consensus decision module evaluates the measurement outcomes produced by the quantum circuit. The six measurement results are represented as

$$\mathbf{Y} = [y_1, y_2, y_3, y_4, y_5, y_6]$$

The XOR operation is then applied to the six values to calculate the measured parity:

$$\mathbf{P}_{\text{measured}} = y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6$$

The resulting parity value is compared with the expected parity generated according to the validation condition. If the measured parity and expected parity satisfy the required condition, the transaction passes the quantum validation stage and can proceed to block creation.

For example, if the measurement output is:

$$\mathbf{Y} = [1, 1, 0, 0, 1, 0]$$

then:

$$1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

Therefore, the measured parity is **1**. If the expected parity is also **1**, the validation condition is satisfied.

4.2.8 SHARED LEDGER REPLICATION MODULE

The shared ledger replication module maintains the blockchain information across the participating validator nodes. After successful transaction validation and block creation, the resulting block is reflected across the validator ledgers.

The replicated ledger provides a consistent representation of the blockchain state among the participating validators. Each validator can therefore maintain the validated blockchain information as part of the distributed architecture.

The replication stage completes the transaction lifecycle by moving the validated transaction from the validation process into the shared blockchain state.

4.3 CLASSICAL BLOCKCHAIN LAYER

The classical blockchain layer provides the foundation of the proposed hybrid architecture. Although the project introduces a quantum validation mechanism, the core blockchain operations remain classical. This layer is responsible for managing transaction information, generating cryptographic hashes, creating blocks, maintaining block relationships, and updating the shared ledger.

The use of a classical blockchain layer ensures that the proposed system retains the fundamental structure of blockchain technology. The quantum mechanism is introduced as an additional validation stage rather than as a complete replacement for the classical blockchain infrastructure.

4.3.1 TRANSACTION AND MEMPOOL PROCESSING

When a user submits a transaction, the transaction is received by the application and prepared for validation. The transaction is temporarily maintained as part of the transaction-processing workflow before it is incorporated into a finalized block.

The transaction information is processed by the classical blockchain component and prepared for cryptographic hashing. The transaction subsequently enters the validation process involving the validator network and quantum validation layer.

4.3.2 SHA-256 HASHING

SHA-256 is used as the cryptographic hashing mechanism within the classical blockchain layer. The hashing process converts the relevant transaction or block information into a fixed-length hash value.

The generated hash is used as part of the blockchain structure and contributes to the linkage between consecutive blocks. If the underlying input data changes, the resulting hash value also changes, allowing modifications to blockchain information to be detected.

4.3.3 BLOCK FORMATION

After successful validation, the transaction is incorporated into a blockchain block. The block contains the required transaction information together with the information needed to establish its relationship with the previous block.

The block formation stage therefore occurs only after the transaction has successfully passed the required validation process. The resulting block becomes part of the blockchain and is subsequently reflected across the validator ledgers.

4.3.4 BLOCK CHAINING

Block chaining maintains the sequential relationship between the blocks in the blockchain. The cryptographic hash associated with the previous block is used as part of the information linking the newly created block to the existing chain.

This structure allows the blockchain to maintain an ordered history of validated transactions. Any modification to information that affects a block's hash can consequently affect the relationship between that block and subsequent blocks.

4.4 QUANTUM VALIDATION LAYER

The quantum validation layer is the primary distinguishing component of the proposed system. It introduces quantum-state preparation and measurement into the classical blockchain transaction lifecycle.

The layer consists of the six-qubit GHZ-state preparation, measurement-angle generation, quantum measurement, measurement-result processing, and parity-based validation. The resulting validation decision is then returned to the classical blockchain layer.

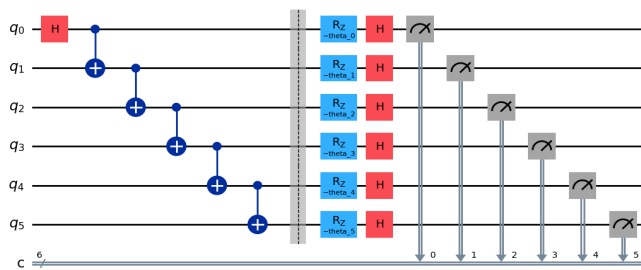


FIGURE 4.1 – QUANTUM CIRCUIT DIAGRAM

4.4.1 QUBIT AND QUANTUM CIRCUIT MODEL

The quantum validation process uses six qubits to represent the six participating validator positions. A qubit is the basic unit of quantum information and can exist in a quantum state that is different from a conventional classical binary state.

The six qubits are initialized and manipulated through quantum circuit operations to produce the required GHZ state. Once the state has been prepared, the measurement operations are applied according to the protocol-defined settings.

The circuit execution produces classical measurement outcomes that are then passed to the parity calculation stage.

4.4.2 SIX-QUBIT GHZ STATE

The six-qubit GHZ state is used to establish a correlated quantum state across the six participating qubits. The state is represented as

$$|\text{GHZ}_6\rangle = 1/\sqrt{2} (|000000\rangle + |111111\rangle)$$

The state provides the quantum correlation required by the proposed validation protocol. The six qubits are subsequently measured using the appropriate measurement settings.

The use of the GHZ state connects the six-validator classical network with the six-qubit quantum validation circuit.

4.4.3 MEASUREMENT ANGLE GENERATION

The measurement-angle generation stage determines the measurement settings used for the six quantum subsystems. The angles are generated according to the requirements of the θ -protocol, with the required relationship between the measurement settings being maintained.

The generated angles determine the measurement basis applied to each participating qubit. These measurement operations produce the binary outcomes that are subsequently used for parity calculation.

4.4.4 QUANTUM MEASUREMENT

After the GHZ state has been prepared and the measurement settings have been assigned, each qubit is measured according to its corresponding setting. The measurement process converts the quantum information into classical binary outcomes.

The six measurement results are collected as a vector:

$$Y = [y_1, y_2, y_3, y_4, y_5, y_6]$$

These values are then passed to the consensus decision module for XOR parity calculation.

4.5 θ -PROTOCOL CONSENSUS MECHANISM

The θ -protocol provides the validation mechanism through which the measurement outcomes of the six-qubit GHZ state are evaluated. The protocol associates measurement settings with the participating quantum subsystems and defines the expected parity relationship for the resulting measurement outcomes.

After measurement, the binary outputs are combined using XOR. The measured parity is calculated as

$$\mathbf{P}_{\text{ca}} = \bigoplus_{i=1}^6 y_i$$

where y_i represents the measurement outcome of the i^{th} qubit.

The calculated parity is then compared with the expected parity. A matching result indicates that the required quantum validation condition has been satisfied.

This process provides the connection between the quantum circuit and the blockchain consensus workflow. The result of the quantum validation is used by the classical system to determine whether the transaction can proceed to block creation.

4.6 MATHEMATICAL FORMULATION

The proposed quantum validation mechanism can be represented mathematically using the six measurement outcomes obtained from the quantum circuit. Let the measurement vector be represented as

$$\mathbf{Y} = [y_1, y_2, y_3, y_4, y_5, y_6]$$

where

$$y_i \in \{0, 1\}$$

for $i = 1, 2, \dots, 6$.

The measured parity is calculated using the XOR operation:

$$\mathbf{P}_{\text{ca}} = y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6$$

The validation condition can then be represented as:

$$\mathbf{P}_{\text{ca}} = \mathbf{P}_{\text{ex}}$$

where $\mathbf{P}_{\text{ex}}$ represents the parity expected according to the defined θ -protocol condition.

If

$$\mathbf{P}_{\text{ca}} = \mathbf{P}_{\text{ex}}$$

then the quantum validation condition is satisfied, and the transaction can proceed to the block creation stage.

then the quantum validation condition is satisfied. Otherwise, the condition is not satisfied.

The mathematical formulation provides a simple representation of the decision process used by the proposed quantum validation component. The actual experimental results obtained from repeated executions of the circuit are discussed in Chapter 5.

4.7 ALGORITHM AND PSEUDOCODE

The complete methodology can be represented as an algorithm that combines classical transaction processing with quantum validation.

Algorithm: Hybrid Classical–Quantum Blockchain Validation

Input: User transaction TT

Output: Validation result and blockchain state

1. Receive transaction T from the user.
2. Process transaction T using the classical blockchain layer.
3. Generate the required SHA-256 hash.
4. Forward the transaction to the validator network.
5. Initialize six quantum qubits.
6. Prepare the six-qubit GHZ state.
7. Generate the measurement angles according to the θ -protocol.
8. Apply the required measurement operations.
9. Measure the six qubits.
10. Collect the binary outputs Y.
11. Calculate the XOR parity of Y.
12. Determine the expected parity.
13. Compare measured parity with expected parity.
14. If the validation condition is satisfied:
 - Create the blockchain block.
 - Link the block with the previous block.
 - Replicate the block across validator ledgers.
 - Return successful validation.
15. Otherwise:
 - Reject or stop the transaction at the validation stage.
16. End.

The algorithm demonstrates the hybrid nature of the proposed system. The initial and final stages are handled by the classical blockchain layer, while the quantum circuit is introduced between transaction

processing and final block creation. This structure allows the quantum validation mechanism to operate as an additional component within the blockchain transaction lifecycle.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION

The proposed hybrid classical–quantum blockchain system was implemented as a software-based prototype that combines conventional blockchain operations with a quantum validation mechanism. The implementation integrates transaction processing, SHA-256 based hashing, block formation, validator participation, quantum measurement, parity verification, consensus decision, and shared ledger representation within a single system. The objective of the implementation was to demonstrate how a quantum-assisted validation layer can be incorporated into a classical blockchain workflow without replacing the fundamental blockchain structures such as transactions, blocks, hashes, previous-hash relationships, and replicated validator records.

The system was implemented using a six-validator network in which each validator is associated with one qubit during the quantum validation process. The quantum component was executed in a simulated environment rather than on physical quantum hardware. The implementation therefore provides a controlled environment for evaluating the proposed validation mechanism and observing the relationship between the generated quantum measurement results and the final consensus decision.

The overall implementation follows the sequence of transaction generation, transaction hashing, validator assignment, quantum validation, consensus verification, block creation, and ledger representation. When a transaction satisfies the required validation condition, it is included in a block. The resulting block maintains its own hash and the hash of the previous block, thereby preserving the classical chain structure. The quantum validation result is additionally associated with the block to indicate whether the validator network successfully satisfied the required quantum consensus condition.

5.2 IMPLEMENTATION ENVIRONMENT

The implementation environment consists of the software components required for classical blockchain processing, quantum circuit simulation, and the web-based presentation of blockchain information. The classical layer handles transaction information, cryptographic hashing, block generation, block chaining, and validator records, while the quantum layer handles qubit preparation, GHZ-state generation, measurement operations, and parity verification.

The quantum validation mechanism was evaluated using a simulator-based environment. This approach allows the quantum circuit and its measurement outcomes to be executed repeatedly under controlled conditions. The use of simulation is particularly suitable for the prototype because the

objective is to evaluate the proposed validation logic and its integration with blockchain consensus rather than to demonstrate execution on physical quantum hardware.

Table 5.1 Implementation Environment

Component	Implementation / Purpose
Blockchain Layer	Classical blockchain transaction and block processing
Hashing	SHA-256
Validator Network	Six validators
Quantum Model	Six-qubit quantum circuit
Quantum State	Six-qubit GHZ state
Quantum Execution	Quantum simulator
Validation Method	Measurement and XOR parity verification
Ledger Model	Replicated validator ledger representation
User Interface	Web-based blockchain monitoring interface

5.3 TRANSACTION PROCESSING IMPLEMENTATION

The transaction processing stage is responsible for receiving and representing the information that is to be recorded on the blockchain. A transaction contains information associated with the sender, receiver, transaction amount, transaction identifier, and transaction hash. Before the transaction can become part of a block, it is subjected to the validation procedure involving the validator network.

The implemented interface demonstrates transactions using blockchain-style addresses and transaction identifiers. For example, the displayed transaction information contains a sender address, receiver address, transaction amount, transaction fee, and validation status. In the demonstrated blockchain instance, a transaction with an amount of **250 QTC** and a **1 QTC fee** is shown with an **APPROVED** validation status. This demonstrates the connection between the transaction layer and the subsequent block formation process.

The transaction hash provides a unique cryptographic representation of the transaction data. The generated hash is subsequently associated with the block containing the transaction. This ensures that the transaction can be represented within the classical blockchain structure while the quantum validation result is maintained as an additional consensus-related property.

5.4 QUANTUM VALIDATION IMPLEMENTATION

The quantum validation component forms the central part of the proposed hybrid system. Six validators are represented by six qubits, with each validator contributing a measurement result to the consensus process. The quantum circuit prepares the required entangled state and applies the measurement operations corresponding to the generated measurement parameters.

A six-qubit GHZ state is used to establish the required quantum correlation among the six validator qubits. The GHZ state allows the measurement results of the participating qubits to be evaluated collectively rather than treating each validator independently. The resulting binary measurement values are then used by the consensus mechanism to determine whether the required parity condition has been satisfied.

The implementation associates each validator with a corresponding qubit and measurement result. The interface demonstrates the six-validator arrangement using the identifiers Validator-1 through Validator-6. This provides a direct representation of the relationship between the validator network and the quantum circuit.

5.5 QUANTUM MEASUREMENT AND PARITY VALIDATION

After the quantum circuit is executed, each of the six validator qubits produces a binary measurement result. These results are represented as a sequence of zeros and ones. The measurement output is then processed using the XOR operation to calculate the overall parity of the six-validator measurement result.

The parity calculation can be represented mathematically as:

$$\mathbf{P} = \mathbf{Y}_1 \oplus \mathbf{Y}_2 \oplus \mathbf{Y}_3 \oplus \mathbf{Y}_4 \oplus \mathbf{Y}_5 \oplus \mathbf{Y}_6$$

where $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_6$ represent the binary measurement results obtained from the six validator qubits and $\mathbf{P}$ represents the measured parity.

The measured parity is compared with the expected parity obtained from the specified validation condition. When the measured parity agrees with the expected parity, the quantum validation is considered successful. The successful result allows the transaction to proceed to block formation. If the required parity condition is not satisfied, the validation process does not produce an accepted consensus result.

In the demonstrated implementation, the validator interface displays individual quantum measurement values for all six validators. For example, the displayed result contains the validator measurements $\mathbf{Y}_1 = 1, \mathbf{Y}_2 = 1, \mathbf{Y}_3 = 0, \mathbf{Y}_4 = 0, \mathbf{Y}_5 = 1, \text{ and } \mathbf{Y}_6 = 0$. These measurements are collectively evaluated by the parity-based validation mechanism.

Table 5.2 Sample Quantum Validation Result

Validator	Measurement Result
Validator-1	1
Validator-2	1
Validator-3	0
Validator-4	0
Validator-5	1
Validator-6	0

The six measurement values are not interpreted individually as independent approval decisions. Instead, they are collectively processed by the XOR parity mechanism. This is important because the proposed quantum validation mechanism evaluates the correlation condition across the complete validator set.

5.6 BLOCK CREATION AND BLOCKCHAIN IMPLEMENTATION

Once the transaction successfully passes the quantum validation condition, it is incorporated into the classical blockchain layer. The generated block contains the transaction information together with cryptographic information required to maintain the chain. Each block contains its own block hash and the hash of the previous block, allowing the blocks to be linked sequentially.

The implementation begins with a genesis block, which acts as the initial block of the chain. Subsequent blocks contain the hash of their immediate predecessor as the previous-hash value. Consequently, modification of an earlier block would affect the hash relationship of subsequent blocks.

The implemented interface demonstrates this relationship through the displayed **Genesis Block #0** and the subsequent **Block #1**. The genesis block contains the initial previous-hash representation, while Block #1 references the hash of the genesis block. This demonstrates the operation of the classical block-chaining mechanism within the proposed hybrid architecture.

The block interface also indicates the proposer responsible for creating the block and displays the quantum consensus status. In the demonstrated result, Block #1 is associated with **Validator-2 (Beta)** as the proposer and displays “**Quantum Consensus: Passed**”, indicating that the block was created following successful quantum validation.

Figure 5.1: Blockchain Block and Quantum Consensus Status

The implementation interface also presents the relationship between the genesis block and the newly created block. The displayed block information includes the block hash, previous hash, transaction hash, transaction count, quantum consensus status, and quantum measurements associated with the validators. This provides a visual representation of how classical blockchain information and quantum validation information are combined within the prototype.

5.7 SHARED LEDGER IMPLEMENTATION

The blockchain network is designed around multiple validators that maintain a representation of the shared ledger. After successful validation and block creation, the resulting block can be represented across the participating validator nodes. The purpose of this arrangement is to demonstrate the distributed nature of blockchain storage while incorporating the quantum validation result into the block-processing workflow.

The implemented interface represents six validators using the identifiers Alpha, Beta, Gamma, Delta, Epsilon, and Zeta. Each validator is associated with a corresponding qubit during the quantum validation process. The validator information displayed for a block includes the validator identifier, qubit identifier, measurement angle, and binary measurement result.

The block details interface therefore provides both classical blockchain information and validator-level quantum information. This combination allows the user to inspect not only the transaction and block hashes but also the validation information produced by the six-validator quantum mechanism.

5.8 WEBSITE IMPLEMENTATION AND RESULTS

The web interface was developed to provide a visual representation of the blockchain network and its validation process. Instead of displaying only raw program output, the interface presents blockchain information in a structured form so that transactions, blocks, validator results, and quantum consensus information can be inspected.

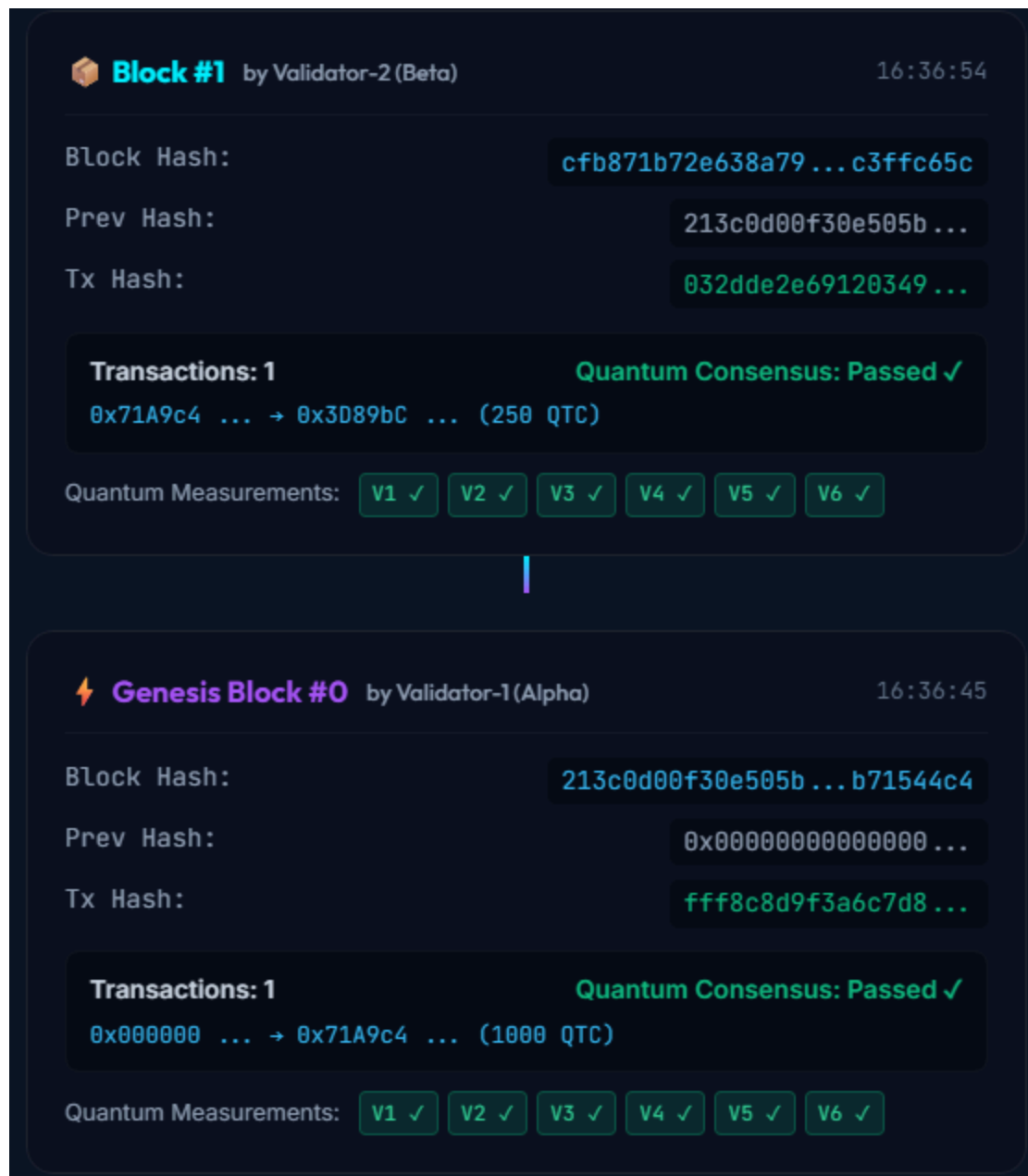


Figure 5.1: Block and Validator Network Display

The first implementation screenshot presents the relationship between the genesis block and the newly generated block. The interface displays **Genesis Block #0** followed by **Block #1**. Each block contains information such as the block hash, previous hash, transaction hash, transaction count, and quantum consensus status. The interface also displays the six quantum measurements as individual validator results.

The displayed Block #1 shows **Quantum Consensus: Passed**, together with the six validator measurement indicators. This demonstrates that the quantum validation stage has been successfully integrated with the classical block creation stage.

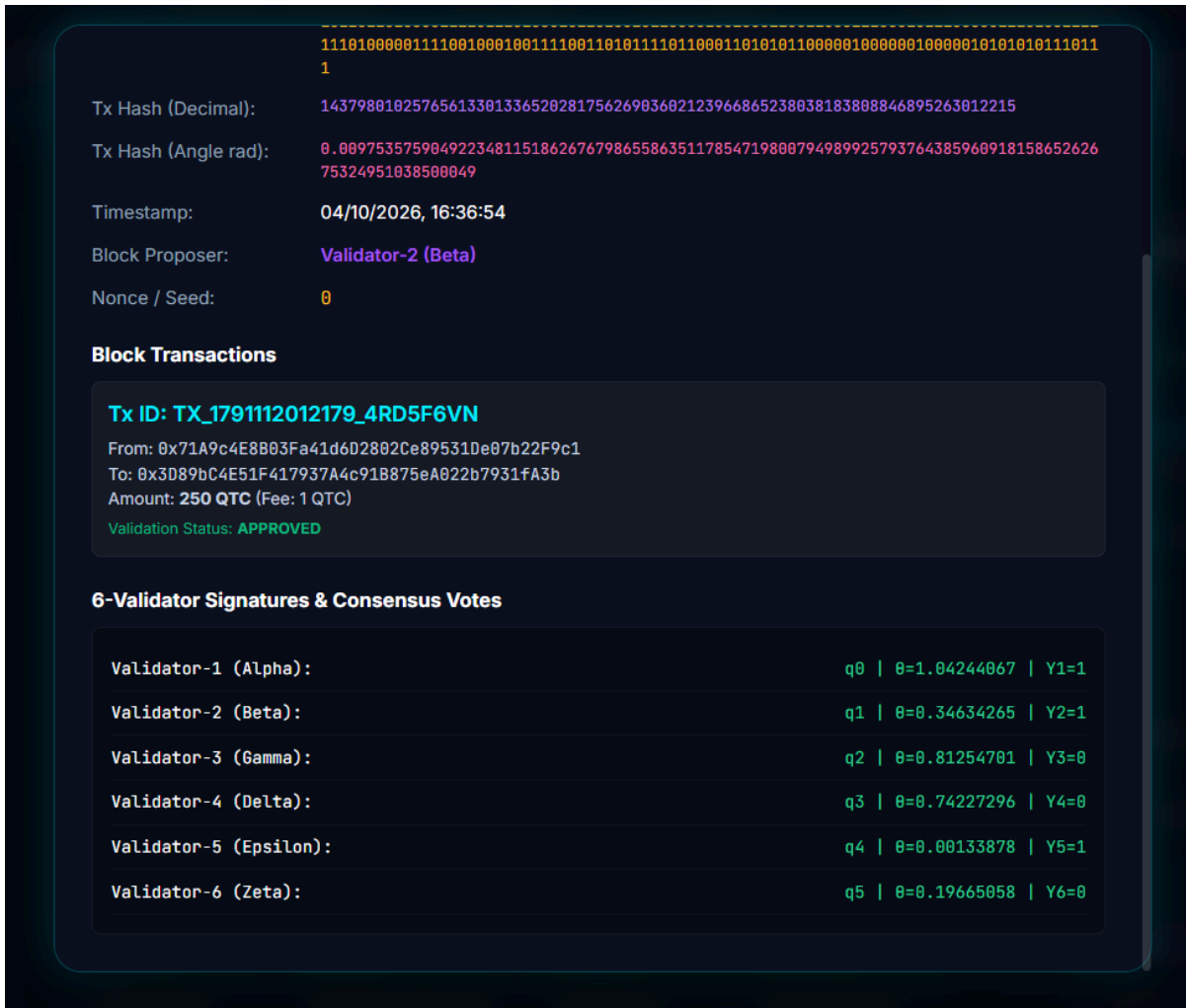


Figure 5.2: Block Transaction and Validator Details

The second implementation screenshot presents detailed information associated with a blockchain block. It includes the transaction hash in multiple representations, the timestamp, the block proposer, the nonce or seed value, and the transaction information contained within the block.

The transaction section displays a transaction identifier together with the sender address, receiver address, transaction amount, and validation status. The demonstrated transaction has an amount of **250 QTC**, a **1 QTC transaction fee**, and an **APPROVED** validation status. The same interface also presents the six-validator consensus information, showing the qubit assigned to each validator, the corresponding measurement angle, and the binary measurement result.

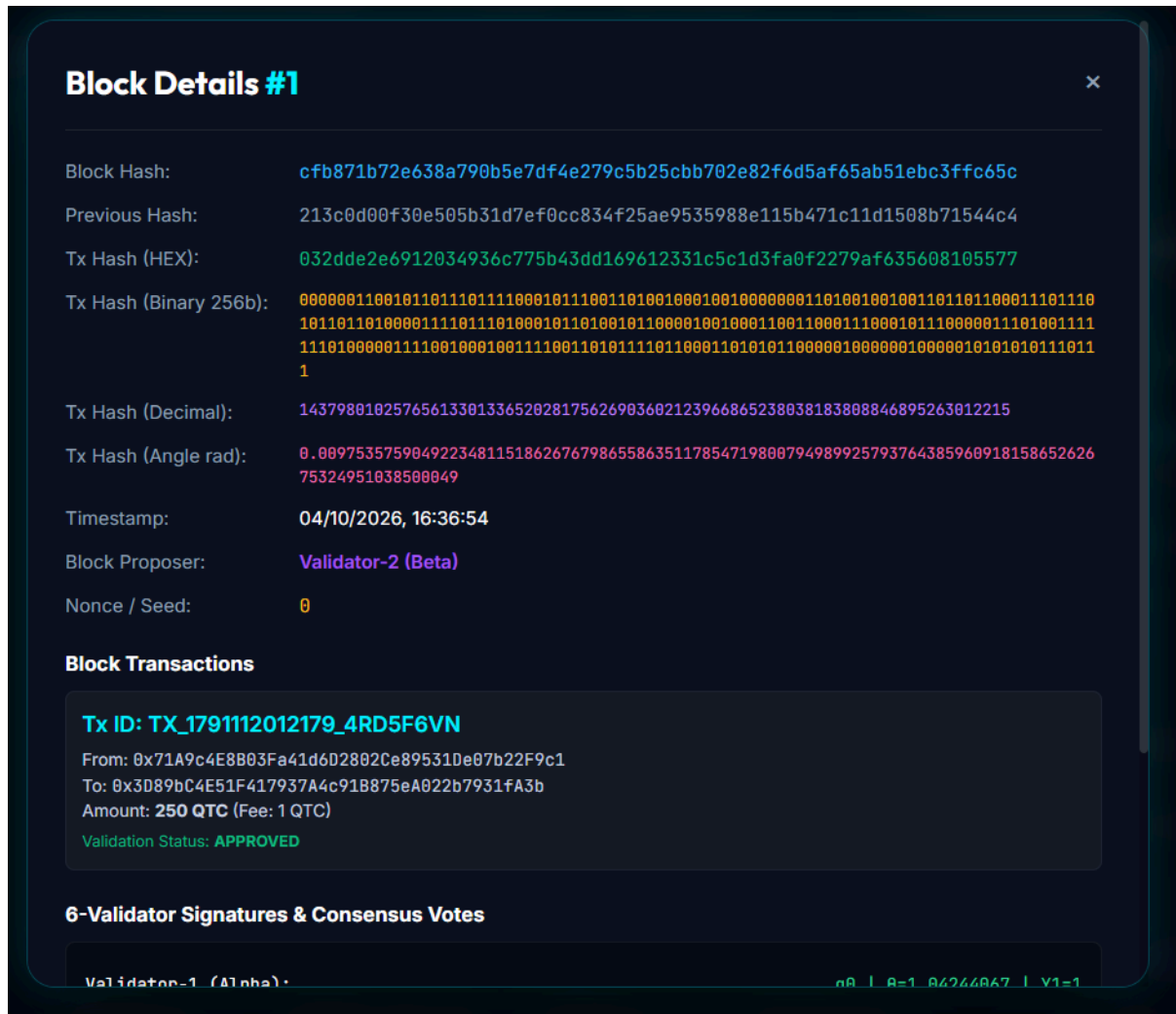


Figure 5.3: Detailed Block Information

The third implementation screenshot provides a detailed view of the generated block. It presents the block hash, previous hash, transaction hash in hexadecimal and binary representations, decimal and angle representations, timestamp, block proposer, and nonce or seed.

The detailed block view demonstrates that the classical cryptographic information and quantum validation information can be inspected together. The block is associated with Validator-2 (Beta) as the proposer, while the transaction is displayed with an approved validation status. The lower section of the interface provides the six-validator signature and consensus information, allowing the validation process to be examined at the individual validator level.

5.9 EXPERIMENTAL RESULTS

The quantum validation mechanism was evaluated through repeated independent trials using the ideal quantum simulation environment. A total of **20 independent quantum-validation trials** were

performed. The experimental evaluation focused on whether the generated measurement results satisfied the required parity condition.

The results obtained from the experiments showed **100% parity validation** under the ideal simulator environment. This indicates that all evaluated trials produced measurement outcomes satisfying the specified quantum correlation and parity condition.

The average execution time recorded for a quantum-validation trial was approximately **3.25 ± 0.05 seconds**. This measurement represents the execution time observed for the simulated validation process and should be interpreted as simulator-based performance rather than physical quantum-hardware execution time.

Table 5.3 Quantum Validation Experimental Results

Evaluation Parameter	Result
Number of independent trials	20
Successful parity validations	20
Parity validation rate	100%
Average execution time	3.25 ± 0.05 seconds/trial
Execution environment	Ideal quantum simulator

The results demonstrate that the proposed six-qubit validation mechanism successfully reproduced the expected parity condition across the evaluated trials. The experiment therefore provides evidence that the quantum validation logic can operate consistently within the simulated environment used for the prototype.

5.10 PERFORMANCE ANALYSIS

The performance evaluation considers the consistency of quantum validation and the execution time of the simulated quantum component. Since the prototype uses a simulator rather than physical quantum hardware, the measured execution time includes the computational overhead associated with simulation. Therefore, the observed timing should not be directly interpreted as the expected execution time of an equivalent circuit on future quantum hardware.

The validation accuracy observed during the experimental trials was 100% under the ideal simulator environment. This demonstrates the reliability of the implemented parity-validation mechanism under the conditions used for the experiment. The average execution time of 3.25 ± 0.05 seconds per trial provides a baseline for the current software implementation.

Table 5.4 Performance Metrics

Metric	Observed Result
Quantum validation trials	20
Successful validation	20/20
Validation rate	100%
Average validation execution time	3.25 ± 0.05 seconds
Validator count	6
Qubit count	6

The performance results demonstrate the functional feasibility of integrating the quantum validation layer with the classical blockchain workflow. However, the results should be considered within the scope of a simulator-based prototype and should not be interpreted as a benchmark for large-scale blockchain deployment.

5.11 RESULT DISCUSSION

The implementation demonstrates that classical blockchain mechanisms and quantum validation operations can be combined within a single software architecture. The classical layer continues to perform the fundamental blockchain functions, including transaction representation, SHA-256 hashing, block creation, previous-hash linking, and ledger representation. The quantum layer operates as an additional validation mechanism involving six validators and six qubits.

One important result of the implementation is the successful connection between the quantum validation outcome and block creation. When the required parity condition is satisfied, the interface identifies the quantum consensus as passed and the corresponding transaction can be represented within the generated block. This demonstrates the intended interaction between the quantum and classical components of the proposed system.

The detailed block interface further demonstrates that the system can retain both classical and quantum-related information. Classical information such as block hashes, transaction hashes, timestamps, and previous hashes is displayed together with validator identifiers, qubit assignments, measurement angles, and measurement results. This provides transparency into the validation process and makes the prototype suitable for demonstrating the proposed hybrid architecture.

The experimental results further indicate that the parity validation mechanism operated consistently during the twenty independent simulator trials. The 100% validation rate confirms that the implemented circuit produced the expected parity condition under the ideal simulation environment.

However, the quantum validation mechanism should not be interpreted as directly authenticating every field of the classical blockchain block. Its primary role in the proposed architecture is to verify the specified GHZ-based quantum correlation and parity condition among the validator measurements. The classical blockchain mechanisms remain responsible for cryptographic hashing and block-chain integrity.

5.12 ADVANTAGES OF THE IMPLEMENTED SYSTEM

The implemented prototype provides an integrated environment for demonstrating classical blockchain and quantum validation concepts. The use of a six-validator network establishes a clear relationship between the validator nodes and the six-qubit quantum circuit. The GHZ-based validation mechanism enables the validator measurements to be evaluated collectively through the parity condition.

Another advantage is the combination of cryptographic blockchain mechanisms with quantum validation. SHA-256 hashing and previous-hash linking preserve the conventional structure of the blockchain, while the quantum component introduces an additional consensus-validation mechanism. The web interface also improves transparency by presenting the block information, transaction information, validator measurements, and quantum consensus result in a single environment.

The simulator-based implementation additionally makes the system reproducible without requiring access to physical quantum computing hardware. This is useful for evaluating the proposed concept and demonstrating the interaction between the classical and quantum components.

5.13 LIMITATIONS

The current implementation is a prototype and therefore has limitations that should be considered when interpreting the results. The quantum validation mechanism is executed using a simulator rather than physical quantum hardware. Consequently, the measured execution time represents simulator performance and cannot be considered a direct measurement of execution on an actual quantum processor.

The evaluation was also performed using a six-validator configuration. Therefore, the observed results do not establish how the system would behave with a substantially larger validator network or a larger number of qubits. Additional experimentation would be required to evaluate scalability, communication overhead, and performance for larger configurations.

Furthermore, the quantum validation mechanism verifies the specified GHZ correlation and parity condition rather than directly authenticating the complete classical block contents. The classical blockchain layer therefore remains essential for transaction hashing, block integrity, and chain construction.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed project presented the implementation and evaluation of a hybrid classical–quantum blockchain system that combines conventional blockchain mechanisms with a GHZ-based quantum validation protocol. The primary objective of the system was to investigate how a quantum validation layer could be integrated with a classical blockchain architecture while preserving fundamental blockchain operations such as transaction processing, cryptographic hashing, block formation, block chaining, validator participation, and shared ledger representation.

The implemented system uses a six-validator network in which each validator is associated with one qubit in the quantum validation process. A six-qubit GHZ state is generated to establish the required quantum correlation among the participating validators. Measurement operations are then performed on the qubits, and the resulting binary values are evaluated using an XOR-based parity mechanism. The measured parity is compared with the expected parity condition to determine whether the quantum validation requirement has been satisfied.

The classical blockchain component continues to perform the core blockchain operations. Transactions are represented and processed before being incorporated into blocks. SHA-256 hashing is used to generate cryptographic representations, while the previous-hash relationship connects consecutive blocks and preserves the chain structure. Once the required quantum validation condition is satisfied, the validated transaction can be incorporated into the blockchain block and represented within the shared ledger.

The implementation was evaluated using an ideal quantum simulation environment. Twenty independent quantum-validation trials were performed, and all twenty trials satisfied the required parity condition, resulting in a validation rate of 100% under the experimental conditions. The average execution time was measured as approximately 3.25 ± 0.05 seconds per trial. These results demonstrate that the implemented quantum validation mechanism operated consistently within the simulator-based environment.

The developed web interface further demonstrates the integration of the classical and quantum components. It provides information about transactions, blocks, hashes, validator identities, qubit assignments, measurement values, and quantum consensus status. This makes it possible to observe the complete validation and block-generation process through a unified interface.

The project therefore demonstrates the functional feasibility of combining classical blockchain processing with a simulated quantum validation mechanism. However, the current implementation should be considered a prototype rather than a production blockchain system. The quantum component validates the specified GHZ correlation and parity condition among the validators; it does

not by itself provide direct authentication of all classical block contents. Similarly, the experimental performance results are based on quantum simulation and therefore should not be interpreted as measurements of physical quantum-hardware performance.

6.2 FUTURE WORK

The proposed system provides a foundation for further research into hybrid classical–quantum blockchain architectures. One important direction for future development is the implementation of the quantum validation protocol on actual quantum hardware. Executing the six-qubit circuit on a physical quantum processor would allow the effects of hardware noise, gate errors, decoherence, and measurement errors to be studied and compared with the ideal simulator results.

Another area of future work is the expansion of the validator network. The current implementation uses six validators and six qubits to demonstrate the proposed validation mechanism. Future versions can investigate larger validator configurations and study how the quantum circuit, consensus process, execution time, and communication requirements change as the network size increases.

Future work can also investigate the integration of more advanced error-mitigation and error-correction techniques. Since practical quantum hardware is affected by noise, additional mechanisms may be required to maintain reliable validation when the proposed protocol is executed outside an ideal simulation environment. Comparing ideal simulation results with noisy simulation and physical-hardware results would provide a more complete evaluation of the protocol.

The blockchain layer can also be extended to support a more realistic distributed network environment. Future implementations may allow multiple independent validator nodes to communicate over a network and maintain synchronized copies of the ledger rather than representing the validator network primarily within a software prototype. This would provide an opportunity to evaluate communication latency, synchronization, fault tolerance, and consensus performance.

Another potential direction is the development of stronger integration between the quantum validation result and classical block verification. The current quantum mechanism primarily verifies the required GHZ correlation and parity condition. Future research can investigate protocols in which quantum-generated validation information is more closely bound to transaction or block-specific data while preserving the intended security and consensus properties of the system.

The user interface can also be enhanced to provide real-time monitoring of validator states, transaction propagation, block generation, quantum circuit execution, and ledger synchronization. Such improvements would make the system more suitable for experimental demonstrations and educational evaluation.

Finally, future research can compare the proposed hybrid approach with conventional blockchain consensus mechanisms using standardized metrics such as validation latency, computational overhead, communication cost, scalability, fault tolerance, and energy requirements. Such comparative evaluation would provide a stronger basis for determining the practical applications and limitations of quantum-assisted consensus mechanisms.

6.3 CHAPTER SUMMARY

This chapter presented the conclusion and future directions of the proposed hybrid classical–quantum blockchain system. The implementation successfully demonstrated the integration of classical blockchain processing with six-qubit GHZ-based quantum validation in a simulator-based environment. The experimental results showed successful parity validation across the evaluated trials, while the developed web interface provided a visual representation of transactions, blocks, validators, quantum measurements, and consensus results.

The project establishes a foundation for further investigation into quantum-assisted blockchain validation. Future development can focus on physical quantum hardware implementation, larger validator networks, noisy quantum environments, distributed validator communication, stronger block-level integration, and comprehensive performance comparisons with conventional consensus mechanisms.

REFERENCES

- [1] Haber, S., Stornetta, W. S.: How to time-stamp a digital document. *Journal of Cryptology* **3**, 99–111 (1991).
- [2] Zhao, W.: Blockchain technology: Development and prospects. *National Science Review* **6**, 369–373 (2019).
- [3] Shor, P. W.: Algorithms for quantum computation: Discrete logarithms and factoring. In: *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*. IEEE Computer Society Press, (2002).
- [4] Grover, L. K.: A fast quantum mechanical algorithm for database search. In: *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing (STOC '96)*, pp. 212–219. ACM Press, New York, NY, USA (1996).
- [5] Kiktenko, E. O., Pozhar, N. O., Anufriev, M. N., Trushechkin, A. S., Yunusov, R. R., Kurochkin, Y. V., Lvovsky, A. I., Fedorov, A. K.: Quantum-secured blockchain. *Quantum Science and Technology* **3**(1), 035004 (2018).
- [6] Reddy, N. R., Suryadevara, S., Reddy, K. G. R., Umamaheswari, R., Guttula, R., Kotoju, R.: Quantum secured blockchain framework for enhancing post quantum data security. *Scientific Reports* **15**, 31048 (2025).
- [7] Goh, K. W., Khan, B. U. I., Khan, A. R., Putra, D. S., P. P., F. R., Bhuyian, M. A.: Eliminating single points of trust: A hybrid quantum and post-quantum blockchain with distributed key generation. *Scientific Reports* **16**, 1286 (2026).
- [8] Banerjee, S., Mukherjee, A., Panigrahi, P. K.: Quantum blockchain using weighted hypergraph states. *Physical Review Research* **2**(1) (2020). <https://doi.org/10.1103/PhysRevResearch.2.013322>

[9] Rajan, D., Visser, M.: Quantum blockchain using entanglement in time. *Quantum Reports* **1**, 3–11 (2019).

[10] Konara, R., Zoysa, K. D., Mahasinghe, A., Sayakkara, A., Ranasinghe, N.: Time-entangled quantum blockchain with phase encoding for classical data. *Quantum Reports* **8**, 69 (2026).

APPENDIX

APPENDIX A – SYSTEM IMPLEMENTATION DETAILS

The Hybrid Classical-Quantum Blockchain System is implemented as a modular software platform that integrates classical blockchain processing with quantum-based validation. The implementation consists of transaction processing, SHA-256 hashing, mempool management, validator selection, quantum circuit execution, parity-based validation, block generation, and shared ledger synchronization.

The system uses six validator nodes, with each validator associated with a corresponding qubit during the quantum validation process. The quantum validation component uses Qiskit and Qiskit Aer to prepare and execute the required quantum circuit in a simulated environment. The resulting binary measurement outcomes are processed collectively to determine the validation status of the transaction.

APPENDIX B – QUANTUM VALIDATION PROCESS

The quantum validation process begins by preparing a six-qubit GHZ state representing the shared quantum state of the six validators.

$$|\text{GHZ}_6\rangle = 1/\sqrt{2} (|000000\rangle + |111111\rangle)$$

The quantum circuit applies the required measurement operations to the six qubits. Each qubit produces a binary measurement outcome represented by:

$$\mathbf{Y} = [y_1, y_2, y_3, y_4, y_5, y_6]$$

where:

$$y_i \in \{0, 1\}$$

for $i = 1, 2, \dots, 6$.

The measurement results are combined using the XOR operation to calculate the measured parity:

$$\mathbf{P}_{\text{measured}} = y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6$$

The measured parity is compared with the expected parity generated according to the defined validation condition. When both values satisfy the required condition, the transaction successfully passes the quantum validation stage and proceeds toward block creation.

APPENDIX C – SAMPLE QUANTUM VALIDATION

A sample measurement output from the six-validator quantum validation process is:

$$Y = [1, 1, 0, 0, 1, 0]$$

The corresponding parity calculation is:

$$1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1$$

Therefore, the measured parity is:

$$P_{\text{measured}} = 1$$

When the expected parity is also:

$$P_{\text{expected}} = 1$$

the validation condition is satisfied and the transaction is permitted to proceed to block formation.

APPENDIX D – SOFTWARE AND IMPLEMENTATION ENVIRONMENT

The software implementation uses Python for the backend and quantum-processing components and JavaScript, HTML5, and CSS for the web-based interface. The quantum execution environment uses Qiskit 2.2.3 and Qiskit Aer 0.17.2, with NumPy 2.4.6 and SciPy 1.17.1 supporting numerical and scientific operations.

The blockchain component implements transaction processing, SHA-256 hashing, mempool handling, validator management, block generation, and ledger synchronization. The quantum component provides GHZ-state preparation, quantum measurement, and parity-based validation.

APPENDIX E – EXPERIMENTAL VALIDATION RESULTS

The quantum validation mechanism was evaluated using twenty independent simulation trials in an ideal noise-free environment. The experimental results demonstrated successful agreement with the required parity condition across the evaluated trials.

Number of Trials: 20

Validation Success Rate: 100%

Average Execution Time: 3.25 ± 0.05 seconds per trial

Number of Validators: 6

Number of Qubits: 6

Execution Environment: Ideal Quantum Simulator

These results demonstrate the functional operation of the implemented quantum validation mechanism within the simulated environment.

APPENDIX F – SYSTEM SCREENSHOTS

The following screenshots are included as evidence of the implemented system and its operation.

Figure A.1: Website Interface – Blockchain and Validator Information

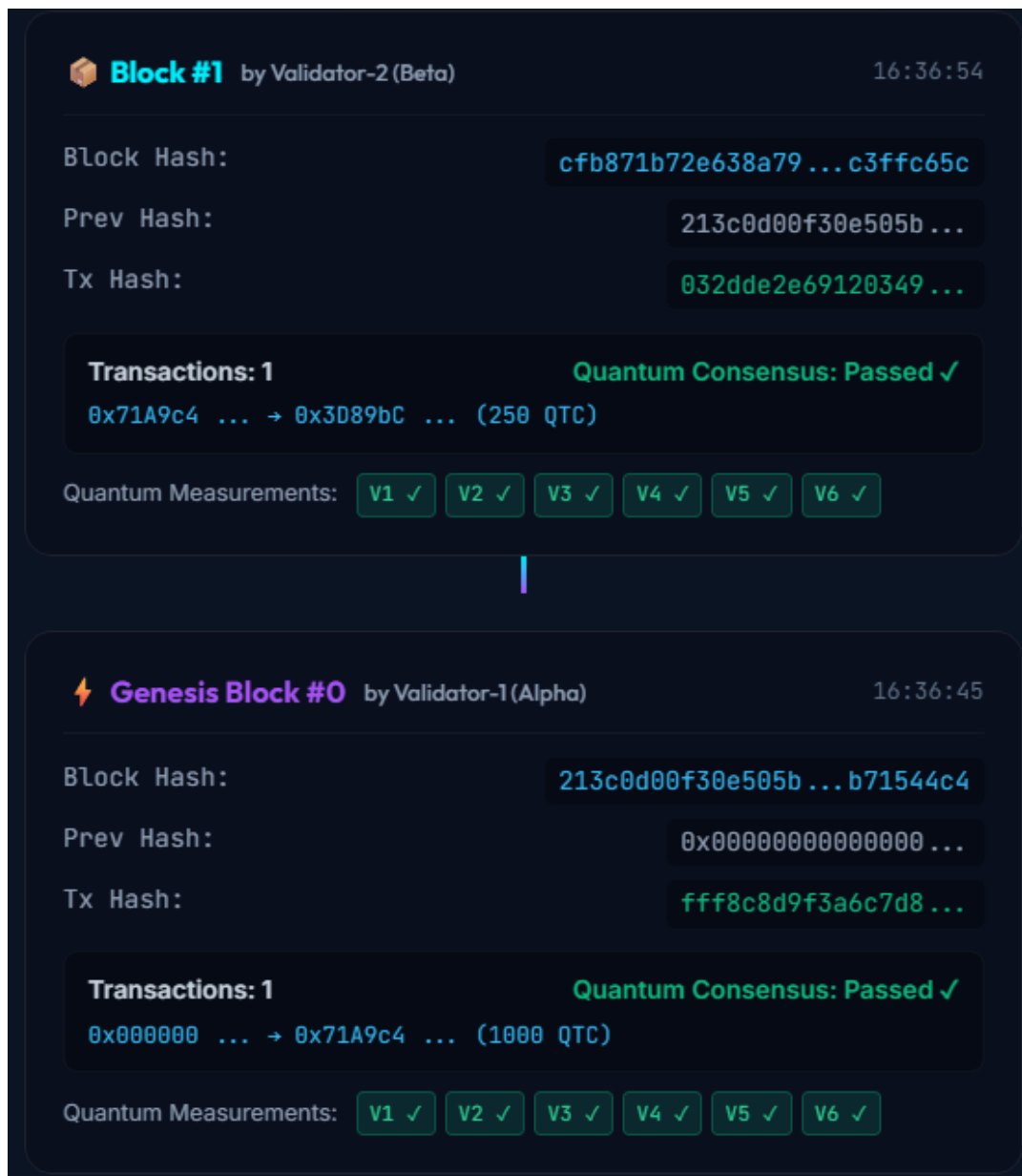


Figure A.2: Website Interface – Transaction and Block Details

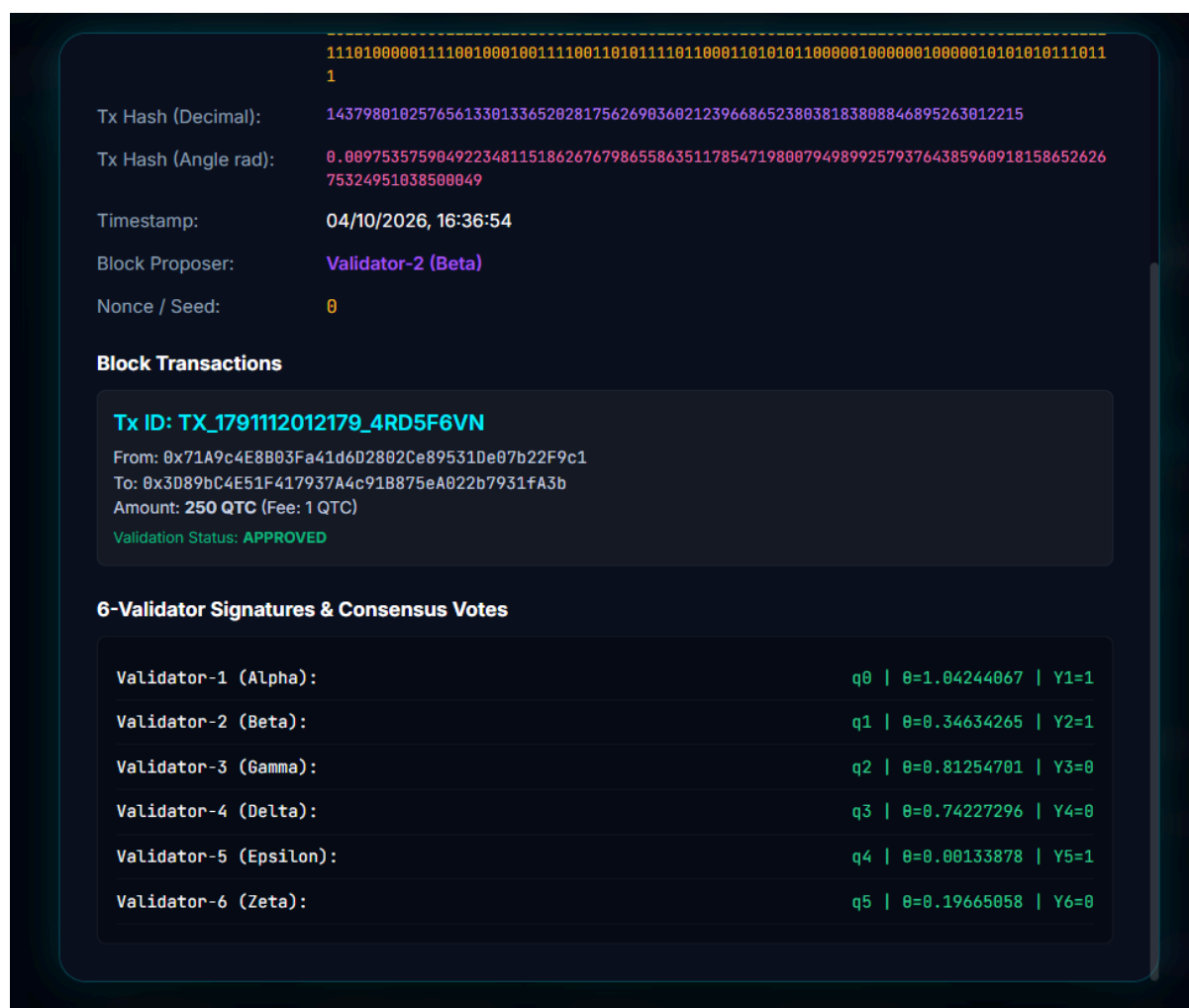


Figure A.3: Website Interface – Quantum Validation and Consensus Details

Block Details #1

Block Hash:

cfb871b72e638a790b5e7df4e279c5b25cbb702e82f6d5af65ab51ebc3ffc65c

Previous Hash:

213c0d00f30e505b31d7ef0cc834f25ae9535988e115b471c11d1508b71544c4

Tx Hash (HEX):

032dde2e6912034936c775b43dd169612331c5c1d3fa0f2279af635608105577

Tx Hash (Binary 256b):

0000001100101101110111100010111001101001000100100000001101001001001101101100011101101101100001111011101000011110111010000111101110100001001000110010001100110001110000111010011111010000011110010001001111001101011110110001101011000001000000100000101010101110111

Tx Hash (Decimal):

1437980102576561330133652028175626903602123966865238038183808846895263012215

Tx Hash (Angle rad):

0.0097535759049223481151862676798655863511785471980079498992579376438596091815865262675324951038500049

Timestamp:

04/10/2026, 16:36:54

Block Proposer:

Validator-2 (Beta)

Nonce / Seed:

0

Block Transactions

Tx ID: TX_1791112012179_4RD5F6VN

From: 0x71A9c4E8B03Fa41d6D2802Ce89531De07b22F9c1

To: 0x3D89bc4E51F417937A4c91B875eA022b7931fA3b

Amount: 250 QTC (Fee: 1 QTC)

Validation Status: APPROVED

6-Validator Signatures & Consensus Votes

Validator-1 (Alpha):

00 | R=1 04244067 | Y1=1

APPENDIX G – KEY IMPLEMENTATION PARAMETERS

Parameter	Value
Number of Validators	6
Number of Qubits	6
Quantum State	Six-Qubit GHZ State
Blockchain Hashing	SHA-256
Quantum Framework	Qiskit 2.2.3
Quantum Simulator	Qiskit Aer 0.17.2

Numerical Library	NumPy 2.4.6
Scientific Library	SciPy 1.17.1
Frontend Technologies	JavaScript, HTML5, CSS
Quantum Validation Method	XOR Parity Validation
Experimental Trials	20
Validation Rate	100%
Average Simulation Time	3.25 ± 0.05 seconds/trial

Implementation and Evaluation of a Hybrid Classical-Quantum Blockchain Using GHZ-Based θ -Protocol Validation

V. Karthick¹, Kasilingam¹, Alfred Sam D.¹, Brijith Manikandan^{1*}

¹Department of Computer Science and Engineering, Rajalakshmi Engineering College, Thandalam, Chennai, Tamil Nadu, India.

*Corresponding author(s). E-mail(s): 230701058@rajalakshmi.edu.in;
Contributing authors: karthick.v@rajalakshmi.edu.in;
230701145@rajalakshmi.edu.in; 230701026@rajalakshmi.edu.in;

1 Abstract

Blockchain systems require a reliable consensus mechanism to establish agreement among distributed participants before transactions are committed to a shared ledger. This work presents a computational implementation of a quantum blockchain validation protocol based on entangled quantum states, integrating the protocol with a classical blockchain transaction and ledger framework. The implementation models six validator nodes as participants in a six-qubit Greenberger–Horne–Zeilinger (GHZ) state. For each candidate transaction, one validator is randomly selected as the verifier, while measurement angles are generated according to the protocol constraint that their sum is an integer multiple of π . Each validator performs the corresponding quantum measurement and returns a binary measurement outcome. Instead of using conventional majority voting or independent approval from validators, the verifier determines consensus from the XOR parity of the collected measurement outcomes and compares it with the parity prescribed by the measurement-angle condition. The implemented system combines this quantum validation stage with classical blockchain operations, including SHA-256 transaction hashing, block generation, validator records, and replication of accepted blocks across a shared ledger. The transaction hash is retained for classical data integrity, while quantum measurement outcomes are used specifically for the consensus decision. The quantum execution layer is separated from the blockchain application through a backend interface, enabling the protocol to be executed using

a Qiskit-based quantum simulator. The resulting system provides a practical proof of concept for reproducing and integrating the quantum validation procedure described in prior work within an executable blockchain environment. It also establishes a software framework that can be extended for experimental evaluation and eventual execution on quantum hardware. Across 20 independent simulation trials, the quantum validation achieved a 100% parity success rate under ideal simulation conditions, with an average execution time of 3.25 ± 0.05 s per trial.

2 Keywords

Quantum Blockchain, GHZ state, Quantum Consensus, Blockchain Validation, Quantum Computing

3 Introduction

Blockchain technology ensures distributed state preservation, in which each of several players contributes to maintaining a common ledger without relying on a single governing authority. In a classical blockchain, there is a combination of different components that act together to achieve coordination. These components include transaction representation, cryptographic hashing, authentication, communication among the participants' nodes, transaction confirmation, achieving consensus, transitioning the state, and maintaining the distributed ledger. When a correct transaction is confirmed, it is put into the block according to the consensus and validity rules, and then the state of the blockchain is updated and preserved by the participants. Thus, the blockchain's reliability is a product of the interactions among different elements of the distributed system rather than the effectiveness of a particular cryptography method.

The principles of blockchain originate from previously existing ideas in area of cryptography and decentralized digital systems. In 1991, researchers Haber and Stornetta presented a solution for cryptographically linking timestamps of digital documents which became the paramount idea of tamper proof series of records [1]. Then there were a number of ventures such as b-money and Bit Gold which were aimed at exploring decentralized digital values and means of recording digital processes without needing the existence of a central authority. The concepts elaborated in the course of these developments later proved to become the building blocks of blockchain techniques and included cryptographic integrity, decentralized activities, distributing records and shared states.

A key change emerged with the release of Satoshi Nakamoto's Bitcoin: A Peer-to-Peer Electronic Cash System [2] and the start of the Bitcoin system in 2009. The introduction of Bitcoin included features like the linking of a hash, digital signatures, peer-to-peer technology, transaction validation, and proof of work making Bitcoin a working decentralized ledger system. Hence, the importance of this development exceeds the limits of the chained blocks allowing us to see how transaction integrity,

communication, consensus, and replicated state can work without a central administrator. After that, blockchain technology developed into something more than just money transfer systems. Ethereum developed in 2013-2014 and was released in 2015, expanded that system by adding a function of smart contracts making blockchain not only a method of transferring money but rather a platform for decentralized applications.

The classical blockchain framework creates the basis upon which quantum blockchain studies have evolved. Specifically, aspects like transaction validity, participation of validators, consensus, state transformation, and maintaining the ledger still apply, even when quantum technology is involved. Nonetheless, quantum computing has added a new feature to the problem of blockchain studies, since quantum computing changes the assumptions regarding the security of classic blockchains and also allows using quantum data as a resource in blockchain systems.

Quantum computing relies on the manipulation of quantum states which have characteristics like superposition, interference, and entanglement. The creation of quantum algorithms has shown that quantum computers could offer computational benefits for certain types of problems. The algorithm proposed by Shor in 1994 proved that integer factorization and discrete logarithms could in principle be solved efficiently on a quantum computer that is sufficiently capable [3]. As these problems form the basis of important public-key cryptographic systems, the finding indicated a possible long-term danger to digital signatures and to other cryptographic methods used in blockchain systems. Grover's algorithm, which was introduced in 1996, showed a quadratic speedup for unstructured search, the complexity being approximately $O(\sqrt{N})$ as opposed to $O(N)$ for a simple classical search [4]. While the implications of Grover's algorithm are different from those of Shor's, it is still relevant to blockchain since hashing and computational search are part of several blockchain security and consensus mechanisms. As a consequence, the tie between quantum computing and blockchain took form along two main lines described above. One direction is defensive and aims at ensuring the safety of existing classical blockchain solutions from future quantum attacks. The other direction is constructive and studies the possibilities of working quantum computation, telecommunication, and quantum mechanics themselves in the realm of blockchain technology. The first direction led to studies of post-quantum blockchain systems and quantum-protected blockchain architecture, while the second direction gave rise to quantum-augmented and purely quantum blockchain technologies. Kiktenko et al. presented one of the first responses to quantum threat by maintaining the classical blockchain structure but changing some components to increase their security. Kiktenko et al. considered quantum-protected blockchain architecture containing Quantum Key Distribution together with an ordinary blockchain setup. The findings showed the feasibility of using Quantum Key Distribution via urban fiber networks, implying that quantum communication can ensure security without the need to turn transactions, blocks, or records into quantum states [5].

Another different course is post-quantum cryptography (PQC) wherein cryptographic primitives that are exposed to quantum attacks have been substituted by those that can withstand the quantum attacks that gave birth to the QuantumShield BC which incorporates quantum random number generation technology, quantum key distribution technology and quantum Byzantine Fault Tolerance [6] as well as utilizing Dual Layer Signature Mechanism [7] Hence, both of their approaches aim to tackle the quantum threat while preserving a basic classical blockchain system. It is essential to note that there is a difference: quantum-secured blockchain uses quantum technology applied to one or more of the relevant security functions whereas post-quantum blockchain changes the cryptographic premises of classical technology regardless of quantum technologies employed.

This development brings us to an important question: instead of using a quantum technology purely for the protection of a classical blockchain, can quantum information itself come into use for representation, validation, or becoming a part of the consensus mechanism of the blockchain? This question represents the emergence of fundamentally quantum or quantum-native blockchain architectures, where quantum state, quantum correlations, phase information, or entanglement will be used to represent information, validate blocks, or decide whether a transition should be accepted.

As a result, there has been considerable growth in various threads of research based on quantum blockchain technology. One such research area deals with the concept of quantum block design, with an emphasis on mapping classical blockchain data onto quantum states. A recent study of Banerjee, Mukherjee, and Panigrahi focused on quantum blockchain technology that is based on quantum hypergraphs and which incorporates phase-based methods for quantum representation of blockchain information. Their research has raised a number of significant questions on the questions of transforming classical data into quantum states and establishing relationships between the subsequent states [8].

A different aspect is instead oriented to achieving quantum consensus and validation based on entanglement. In ordinary blockchains, the nodes that participate in the blockchain can apply their own validation to a transaction and thus validate it through the vote or any form of agreement. The entanglement method is about the question of whether from the correlations existing among the measurements done by quantum systems, evidence can be obtained confirming that the participants have a legitimate quantum state. In this respect, GHZ states are very important because they provide for the correlations involving different quantum systems. Each qubit can be associated with a participating node, and their results can be studied to see whether the measured correlations correlate with the expected one.

Consequently, this indicates a significant shift in the validator's function. In a classical validation process, Transaction→Individual Validation→Vote/Approval, while an entanglement based procedure can be shown as Shared Quantum State→Distributed Measurement→Correlated Outcomes→Quantum Validation Condition. Thus, the

validator can be thought of as not being an independent voter. The validator becomes a participant in a quantum measurement process with the final decision based on the quality of information between measurement outcomes, which is one of the key points of this work. In addition to these quantum-native method, hybrid quantum-classical architecture is being actively developed since the current state of quantum hardware and quantum networking cannot perform all required functions of a classical distributed ledger. In hybrid architecture, classical systems can do everything from managing transaction data, hashing, metadata, networking, application logic, and persistent ledger storage, while quantum resources involved for separate functions like state preparation, entanglement generation, validation, authentication, and signatures. This means that “quantum blockchain” is not a term for a well-defined architecture but may refer to various replacements of the conventional blockchain architecture or its components with quantum technology. In other words, it can be introduced in order to ensure protection against quantum attacks, transmission of quantum messages, creation of quantum data, verification of the states of the system including that it has achieved consensus, etc. Thus, putting a quantum technology into a blockchain system is not a sufficient condition for saying that this system is quantum.

During this development, the work of Del Rajan and Matt Visser, titled Quantum Blockchain Using Entanglement in Time,[9] serves as the basis for this study. The work was first uploaded to arXiv in 2018 and then published in Quantum Reports in 2019. The main issue is that quantum entanglement is applied as one of the building blocks of blockchain but not just as a means of increasing security. The suggested construction is based on the concept of a temporal Greenberger–Horne–Zeilinger (GHZ) state that builds correlations among quantum systems that are related in time and incorporates those correlations into the blockchain structure.

The Rajan-Visser framework is important because it establishes a link between quantum entanglement and the process of blockchain validation. In accordance with the framework, an individual among the nodes engaged in the process is assigned the role of a verifier, and the quantum block is distributed to all the participating nodes. The verifier presents measurement parameters, following the premise of the protocol, and each of the other participating nodes executes the measurement on its own quantum instance. Afterwards, the nodes send a binary result of their measurements to the verifier, and the verifier processes their results according to the XOR/parity condition based on the measurement angles.

The importance of this mechanism rests on three interconnected components: Shared Quantum State→Distributed Measurements→Collective Validation Condition. The shared quantum state creates correlations among the participating network nodes, the distributed measurements provide a method of turning quantum signals into classical information, and the parity condition translates the correlated classical information into a blockchain-level validation.

Simultaneously, the work done by Rajan–Visser provides a conceptual framework for validation rather than an actual application of a traditional blockchain network. The authors indicate the need of development of low-level consensus and quantum network implementation details. This distinction creates a crucial implementation-level background for further work: the conceptual mechanism explains how a quantum correlation can be associated with validation while the engineering problem is how to connect this mechanism to transaction processing, validator’s management, block generation as well as ledger’s status and the real quantum execution.

The current research investigates this implementation and integration layer. Instead of introducing a new entanglement-based consensus method, this research creates an executing hybrid environment of blockchain that is based on the quantum verification process as described in the Rajan-Visser framework. This serves as a foundation for creating an executable workflow of transition from quantum and classic verification processes. The contribution made in the present study can therefore be regarded as an implementation study of a classical quantum verification technique. The proof-of-concept thus created forms grounds for future examinations with regard to quantum noise, multiple measurements, dishonest or defective participants, different validator configurations, scalability, and depositing on a real quantum machine.

4 Methods

A hybrid classical-quantum blockchain prototype was built here. Conventional blockchain tasks like transaction handling, hashing, block assembly, and ledger copying stay unchanged, but validation uses an entanglement-based quantum method. This relies on the Theta-protocol from quantum blockchain studies, especially the GHZ-state check by Rajan and Visser. Six validators act as six people sharing a six-qubit GHZ state. For every new transaction, one validator Picks randomly to verify it, and six measurement angles follow the needed angle rule. One shared GHZ circuit forms, measurements hit all six qubits, giving one bit result per validator. These six bits go back to the blockchain part, where the checker tests the protocol’s XOR parity rule. Only if that rule passes does the transaction move forward to make a block and lock into the shared ledger.

We retain the classical blockchain mechanisms wherever necessary. For example, we use classical mechanisms such as SHA-256 processing on each transaction to generate a deterministic “transaction hash” that supports classical data integrity/transaction identification mechanisms. Our use of the quantum mechanism is limited to the validation (or consensus) stage; we do not replace classical hashing with our quantum approach. In particular, the numeric representations derived from the SHA-256 transaction hash differ significantly from the angle measurements generated via the qubit’s behavior under the Θ -protocol. In other words, we treat the transaction hash as part of the classical blockchain, while ensuring quantum measurement angles can be generated independent of the transaction hash value. These quantum measurement angles

are created based upon the requirements for successful operation of the quantum validation process itself. Thus, we preserve the ability to separately evaluate performance aspects of the classical and quantum components.

At a high level, the classical layer handles aspects such as transaction processing, hashing, block building, and ledger management. Quantum components such as GHZ-state preparation, distributed measurements, correlated outcomes, and their corresponding validation condition are handled by the second layer of this structure. Finally, the actual execution of the quantum component itself — including all the elements of the computation discussed above — is decoupled from the rest of the blockchain application via a dedicated backend interface. This allows one (for example) to use a Qiskit-based quantum simulator that will run the circuit on behalf of the system and provide results (i.e., the measurement outcomes) back to the blockchain layer over an API connection. This allows a modular architecture where subsequent blockchain-level workflows could potentially plug into other quantum backends without having to redo all of the ledger implementation. In this case, the implementation is a computational proof of concept instead of an actual realization of a quantum network. The paper on which the project is based explains the quantum blockchain as a conceptual design that uses GHZ-state verification as a method of block verification.

4.1 Experimental Environment

4.1.1 Computing Platform

The goal of the experiment is to assess the blockchain process and the quantum verification procedure without relying on cloud quantum technology. Therefore, the actual running of the circuit is done locally with the help of Qiskit Aer. In that way, the experimental environment consists of two layers cooperating with each other: Classical Application Layer + Local Quantum Execution Layer. In the classical layer, the transactions, validators, blocks, consensus state, and ledger synchronization are controlled, while the quantum layer develops and runs the six-qubit verification circuit and returns the six measurement data for block acceptance assessment.

4.1.2 Software environment

The principal software components used for the quantum implementation are given in Table 1.

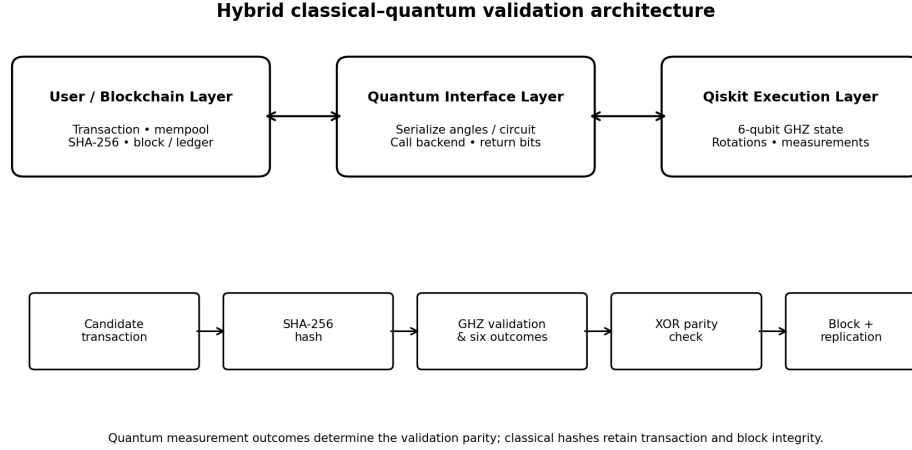
Quantum execution code specifically imports `QuantumCircuit` from Qiskit and `AerSimulator` from Qiskit Aer. The simulator backend accepts six measurement angles, constructs a six-qubit circuit, executes it in one shot, and returns six binary measurement results. Use of local Aer backend is intentional. No IBM Quantum hardware was utilized in the experiment and no cloud execution was needed for the consensus Pipeline. Thus, hardware queue time, cloud-access requirements, and hardware-related noise were removed from the baseline version.

Table 1: Principal components used in the proposed system

Component	Version / Configuration	Role
Python	Python environment used for the Qiskit backend	Quantum execution
Qiskit	2.2.3	Quantum circuit construction
Qiskit Aer	0.17.2	Local quantum simulation
NumPy	2.4.6	Numerical operations
SciPy	1.17.1	Scientific-computing environment
HTML5	–	User-interface layer
CSS	–	User-interface presentation
JavaScript	–	Blockchain application and control logic

4.2 System Architecture

The entire system architecture is comprised of three major layers which is described in the Figure 1. The figure also gives a expanded view of the overall workflow that happens within the 3 layers.

**Fig. 1:** 3 Layer Architecture

The user and blockchain layer consists of the transaction model, mempool, validator nodes, blocks, public ledger, conventional hash functions, and consensus mechanism. The quantum interface layer is responsible for transformation of the blockchain-based measurement parameters to the serialized six-qubit circuit description and invokes the quantum backend. The Qiskit execution layer forms the actual

QuantumCircuit, prepares the GHZ state, performs local angle-dependent measurements, runs the circuit using AerSimulator and obtains six measurement results. This separation is reflected in the architecture of the software.

4.3 Blockchain Network Configuration

As illustrated in Figure 2, our blockchain has one primary user node with six validator nodes (denoted by $V = V1, V2, V3, V4, V5, V6$). These six validators have their unique identifiers, network addresses, states and their own copies of the blockchain ledger. We map these six validator nodes to the six qubits in our quantum verification circuit where $V_i \leftrightarrow q_i$ for $i \in 1, \dots, 6$. Therefore, there are six validators $\equiv$ six quantum participants $\equiv$ six qubits. To emphasize this, note that all the validators are not individual quantum circuits but only part of one large shared quantum circuit – thus, having six quantum participants means they share one big quantum circuit representing the collective verification state. Also, each quantum participant (validator) can give us one outcome of the measurements made on them (Y_i).

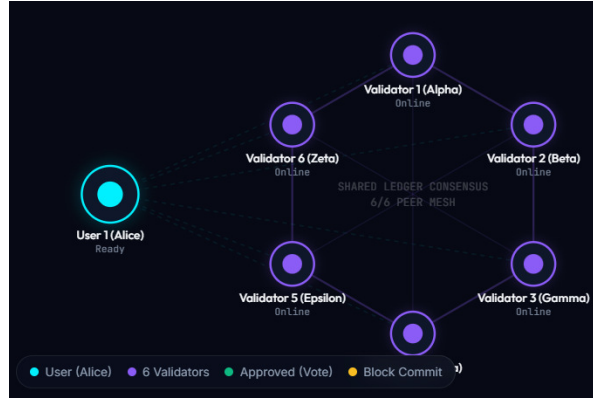


Fig. 2: Blockchain Network Arrangement

Our block chain has various different state structures including Shared Ledger, Mempool and Local Validator Ledgers. The Shared Ledger shows up as an authoritative source of truth held within the application itself. While every validator holds their own synchronized copy of the blockchain's ledger. In order to avoid concurrent processes of more than one transaction we implemented a lock around the process which allows us to process just the very first transaction in the Mempool before moving on to the second.

4.4 Genesis Block Initialization

Prior to starting normal transaction processing, the blockchain needs to be initialized via a genesis block. A genesis transaction gets generated internally, which sets an initial balance to the primary user. Its SHA-256 transaction hash gets computed. Its

status gets marked as committed, and based on that, its corresponding genesis block gets created having zero-value for its previous-hash field. This genesis block will get inserted in the shared ledger and simultaneously replicated across the local ledgers of all validators. Thus, this genesis block is primarily utilized to initialize/establish the initial common states of all six- validator networks. The protocol is such that $B_0 \rightarrow \{L_1, L_2, L_3, L_4, L_5, L_6\}$. (In other words, the quantum Θ -protocol isn't used for creating this initial bootstrapPing state - the quantum validation mechanism will be applied to subsequently submitted candidates.) By having this property, we prevent our initialization phase from requiring an invalid consensus state. To ensure this happens correctly, our implementation explicitly synchronized the genesis block with all 6 validators prior to allowing normal transaction submission.

4.5 Transaction flow

Transaction initiation begins from the perspective of the primary user who submits the transaction containing relevant parameters such as sender, recipient, amount to send, transaction fee, timestamp, metadata, and transaction identifier. Prior to insertion into our validation Pipeline, basic classical checks are run ensuring positivity of the amount parameter and that the sender has sufficient balance to pay for both the amount and transaction fees (in order). Once validated, this transaction will be placed into the mempool which serves as a temporary storage system for transactions which have been created but not yet committed to the blockchain's stateful ledger. For simplicity reasons during our experiments, we've designed the Pipeline to only handle processing of one transaction at a time as shown in Eq 1,

$$T_0 \rightarrow \text{Mempool} \xrightarrow[\text{Classical}]{\text{Quantum}} \text{Block } T_0 \quad (1)$$

Such an approach ensures that we may cleanly associate all results directly attributable to the quantum validation with just the one candidate transaction. Prior to running quantum validation on a transaction, it was required to compute its hash in accordance with SHA-256 producing a 256-bit digest representing the contents of the transaction. In addition to the hexadecimal encoding provided natively by the algorithm, we've recorded the same result using the binary and decimal bases. To avoid losing any bit due to JavaScript rounding issues when storing large integers, the latter is computed via BigInt instead of relying upon the native Number class. It should also be noted that when considering the binary format of the hash, care is taken to maintain the exact 256 bits of output including leading zeros. Finally, we derive a classical transaction associated angle defined as:

$$\Theta_{\text{tx}} = \frac{D}{2^{256} - 1} * \frac{\pi}{4}, 0 \leq \Theta_{\text{tx}} \leq \frac{\pi}{4} \quad (2)$$

From Eq 2, D represents the decimal equivalent value of the given 256-bit transaction hash. This guarantees the resultant angle falls within $[0, \pi/4]$. It should be noted that the value above is never considered part of the set of quantum measurement angles Θ_j participating in the consensus protocol. This distinction ensures that the quantum protocol remains separate from the classical tx_hash representation. Consequently, the transaction-associated parameter differs from the measurement angles, i.e., $\Theta_{\text{tx}} \neq \Theta_j$.

Once the transaction hash has been generated, one of the six validators is randomly selected to act as the verifier for the current validation round. Let the randomly selected validator index be defined as:

$$k = \lfloor 6r \rfloor, \quad r \sim \mathcal{U}(0, 1) \quad (3)$$

Thus, in Eq 3, $k \in \{0, 1, \dots, 5\}$, and validator V_k acts as the verifier for the current transaction. It should also be noted that the remaining validators are not merely independent classical voters; rather, each validator acts as a quantum node associated with its respective qubit of the shared GHZ state.

4.6 Generation of the Six Quantum Measurement Angles

In this case we need six measurement angles. Implementing this configuration you will generate 5 random positive values which are going to be used to define the measure angles, making sure these five angles cover an approximate 95% of the interval associated with π . For the last one we have calculated its value considering the rest of the range.

Let $r_1, r_2, \dots, r_5$ be five random values generated by the implementation, and let

$$R = \sum_{i=1}^5 r_i, \quad \alpha = \frac{0.95}{R} \quad (4)$$

In Eq 4 we have the sum of the 5 random values and a scaling coefficient factor is defined by α .

The first five measurement angles are calculated by Eq 5

$$\Theta_i = \Pi * \alpha * r_i, \quad i = 1, \dots, 5. \quad (5)$$

The final angle is calculated by the Eq 6

$$\Theta_6 = \Pi - \sum_{i=1}^5 \Theta_i, \quad \sum_{i=1}^6 \Theta_i = \Pi \quad (6)$$

We have implemented up to the normal floating point representation. To enforce the six angle constraint we have made this restriction explicit. I.e., the function will reject configurations containing anything but 6 validators. Given that the sum of the six measurement angles is constrained to equal Π , the right-hand side of the Θ -protocol parity condition becomes 1 as defined by Eq 7

$$\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \bmod 2 = \frac{\Pi}{\Pi} \bmod 2 = 1 \quad (7)$$

Therefore, we expect the six-validator experiment to produce an output satisfying the following property:

$$\left(\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \right) \bmod 2 = 1 \quad (8)$$

This rather strange-looking design of Eq 8 leads to an extremely simple final consensus condition of just requiring that the XOR of these six measured binary outcomes equals 1.

4.7 Construction of the Six-Qubit GHZ State

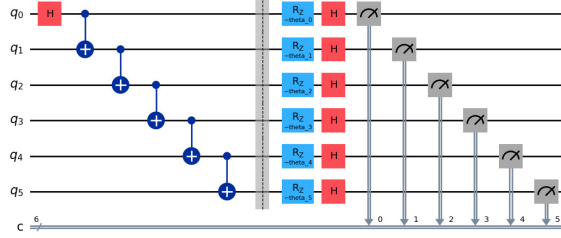


Fig. 3: Qiskit Implementation of 6 qubit GHZ circuit

As in the Figure 3, Quantum validation circuit has six qubits and six classical measurement bits ($q_0, q_1 \dots, q_5$) and ($c_0, c_1 \dots, c_5$). Initially these are set to $|000000\rangle$.

Then applying a Hadamard gate on the qubit q_0 we get Eq 9,

$$|000000\rangle \rightarrow \frac{1}{\sqrt{2}}(|000000\rangle + |100000\rangle) \quad (9)$$

Next applying a sequence of linear CNOT entanglement chain of gates yields the desired six qubit GHZ state which gives Eq 10,

$$|GHZ_6\rangle = \frac{1}{\sqrt{2}}(|000000\rangle + |111111\rangle) \quad (10)$$

Instead of implementing six independent circuits for each participant as would be expected from an ordinary GHZ circuit, it's implemented here as a single quantum circuit that creates a commonly shared quantum state between six validators.

4.8 Final Integration into the classical flow

The quantum-validation stage performs measurements in the Θ basis by applying an $R_z(-\Theta_j)$ operation followed by a Hadamard gate H to each validator qubit, yielding a binary measurement result $Y_j \in \{0, 1\}$.

Once computed, the browser application transmits the six measurement angles and the circuit description to the Python-based quantum backend over the network connection. After processing these inputs, the quantum backend returns the resulting six-bit vector as defined in Eq 11, whose result can be mapped to the corresponding validators.

$$\mathbf{Y} = [Y_1, Y_2, Y_3, Y_4, Y_5, Y_6], \quad V_i \leftrightarrow (q_i, \Theta_i, Y_i) \quad (11)$$

These measured bits represent a single quantum-validation event rather than independent approvals. Accordingly, consensus is calculated using the measured parity by Eq 12, where $\oplus$ denotes the exclusive OR (XOR) operation.

$$P_{\text{measured}} = Y_1 \oplus Y_2 \oplus Y_3 \oplus Y_4 \oplus Y_5 \oplus Y_6 \quad (12)$$

The measured parity is compared against the protocol condition and measurement angle constraint which gives $P_{\text{expected}} = 1$ by Eq 13.

$$P_{\text{expected}} = \left(\frac{1}{\Pi} \sum_{j=1}^6 \Theta_j \right) \bmod 2, \quad \sum_{j=1}^6 \Theta_j = \Pi \quad (13)$$

Therefore, for the transaction to transition from the **PENDING** state to the **VALIDATED** state, the measured parity must satisfy the Eq 14.

$$P_{\text{measured}} = P_{\text{expected}} = 1 \quad (14)$$

Finally, after storing the quantum-verification results along with the transaction data itself within our system, the next step involves generating a classical block consisting of the current transaction, its associated previous block hash, and transaction hash, as well as both the proposer and the records of verification provided by each validator. This last block can subsequently be secured using SHA-256 hashing. Importantly, though, one should note that the role of proposer comes from deterministic selection among those participating as validators; importantly, this position differs from that randomly chosen verifier who participated during the quantum validation/verification process. With this final step completed, the proposed block becomes part of the shared ledger and is distributed to all participants in the six-validator network. Ledger updates propagate across the distributed system by ensuring that each participant receives an identical copy of the validated block before appending it to their respective local ledger.

Following block propagation, user's balances are updated accordingly, and the contents of the transaction mempool are synchronized with the latest ledger state. To maintain system integrity and ensure correct operation, additional validation checks verify the relationships among the previous block hash, the current block hash, and the transaction hashes. These checks help ensure consistency across the distributed ledger and detect discrepancies between local ledger copies.

5 Results

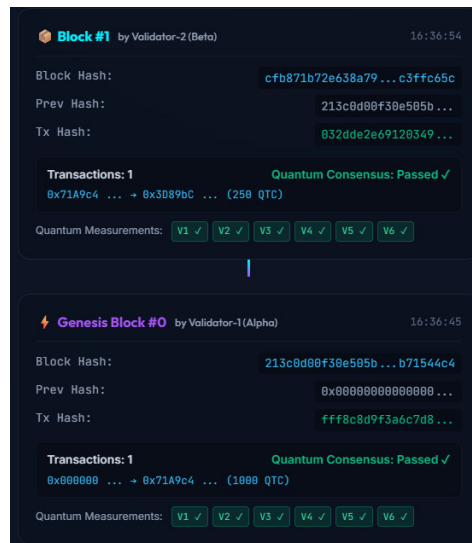


Fig. 4: Shared Ledger

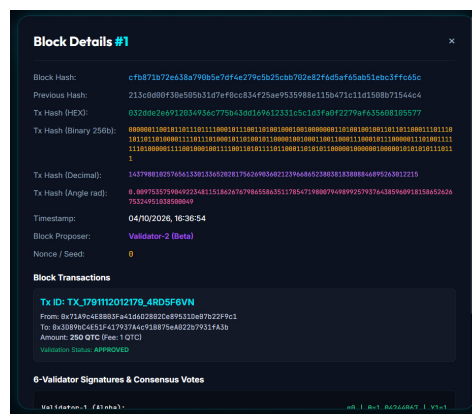


Fig. 5: Block information

based Validation using the above mentioned protocol we found out some constraints which are very crucial from the point of view of actual implementations. First, we see that the Verification condition (collective sum) requires the measurements angles should satisfy certain multiple of π i.e., π . An ideal simulator could ensure this constraint. However, real world Quantum Hardware suffers from finite precision control gates and cumulative angle errors leading to possible deviation in the effective sum j thus reducing the reliability of desired parity output. Second, even though the proposed framework provides us with an encoding mechanism for representing the blockchain record as a entangle quantum state, the proposed Θ -protocol only verify the GHZ correlation of the candidate state, instead of verifying the presence of the intended block information in the state. As per the paper’s discussion they have described about how their classical block records were encoded into temporal Bell and GHZ States, but the part where you test if your supplied candidate State has the GHZ Correlations required is the Verification Stage here. So basically they showed off their Quantum Validation Mechanism. However, to build up a full-fledged Block Authentication Mechanism, we need to explicitly bind the classical block information we want to authenticate with the Candidate State to be Verified. This becomes really important because someone could simply go onto the Classical Blockchain Network, and tamper the block contents while keeping the underlying Quantum Consensus untouched. And since the Quantum Consensus doesn’t care about what happened on the classical side of things, it just happily accepts the tampered block. Now the recent paper regarding Time Entanglements in 2026 [10] manages to overcome this binding challenge, however it runs into serious challenges as the Number Of Blocks Increase, hence working out ways to Combine both methodologies may form an open research problem in our near future scope of Research.

Author contributions

Kasilingam, Alfred Sam D, and Brijith Manikandan contributed equally to the conception, implementation, experimentation, and manuscript drafting. V.Karthick supervised the project, provided technical guidance, and reviewed the final manuscript.

Funding information

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The data that was analyzed and implementation code for user interface and code implemented using QISKIT are available for academic use.

Declarations

Clinical trial number

Not applicable.

Ethics approval and consent to participate

This article does not contain any studies with human participants or animal performed by any of the authors. Hence, ethics approval was not required.

Consent for publication

Not applicable

Competing Interests

The authors declare no competing interests.

References

- [1] Haber, S., Stornetta, W.S.: How to time-stamp a digital document. *Journal of Cryptology* **3**, 99–111 (1991)
- [2] Zhao, W.: Blockchain technology: Development and prospects. *National Science Review* **6**, 369–373 (2019)
- [3] Shor, P.W.: Algorithms for quantum computation: Discrete logarithms and factoring. In: *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*. IEEE Computer Society Press, ??? (2002)
- [4] Grover, L.K.: A fast quantum mechanical algorithm for database search. In: *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing (STOC '96)*, pp. 212–219. ACM Press, New York, NY, USA (1996)
- [5] Kiktenko, E.O., Pozhar, N.O., Anufriev, M.N., Trushechkin, A.S., Yunusov, R.R., Kurochkin, Y.V., Lvovsky, A.I., Fedorov, A.K.: Quantum-secured blockchain. *Quantum Science and Technology* **3**(1), 035004 (2018)
- [6] Reddy, N.R., Suryadevara, S., Reddy, K.G.R., Umamaheswari, R., Guttula, R., Kotoju, R.: Quantum secured blockchain framework for enhancing post quantum data security. *Scientific Reports* **15**, 31048 (2025)
- [7] Goh, K.W., Khan, B.U.I., Khan, A.R., Putra, D.S., P. P., F.R., Bhuyian, M.A.: Eliminating single points of trust: A hybrid quantum and post-quantum blockchain with distributed key generation. *Scientific Reports* **16**, 1286 (2026)
- [8] Banerjee, S., Mukherjee, A., Panigrahi, P.K.: Quantum blockchain using weighted hypergraph states. *Physical Review Research* **2**(1) (2020) <https://doi.org/10.1103/PhysRevResearch.2.013322>
- [9] Rajan, D., Visser, M.: Quantum blockchain using entanglement in time. *Quantum Reports* **1**, 3–11 (2019)

- [10] Konara, R., Zoysa, K.D., Mahasinghe, A., Sayakkara, A., Ranasinghe, N.: Time-entangled quantum blockchain with phase encoding for classical data. *Quantum Reports* **8**, 69 (2026)

Test Author

Journal Final Draft.pdf

 The Northcap University

Document Details

Submission ID

trn:oid::4:583003099

Submission Date

Oct 4, 2026, 6:48 PM GMT

Download Date

Oct 4, 2026, 6:49 PM GMT

File Name

Journal Final Draft.pdf

File Size

297.0 KB

16 Pages**5,803 Words****32,682 Characters**

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Implementation and Evaluation of a Hybrid Classical–Quantum Blockchain Using GHZ Based Θ Protocol Validation

V. Karthick¹

karthick.v@rajalakshmi.edu.in

Kasilingam¹

230701145@rajalakshmi.edu.in

Alfred Sam D.¹

230701026@rajalakshmi.edu.in

Brijith Manikandan^{1,*}

230701058@rajalakshmi.edu.in

Department of Computer Science and Engineering,
Rajalakshmi Engineering College, Thandalam, Chennai, Tamil Nadu, India

*Corresponding author

1 Abstract

Blockchain systems require a reliable consensus mechanism to establish agreement among distributed participants before transactions are committed to a shared ledger. This work presents a computational implementation of a quantum blockchain validation protocol based on entangled quantum states, integrating the protocol with a classical blockchain transaction and ledger framework. The implementation models six validator nodes as participants in a six qubit Greenberger–Horne–Zeilinger (GHZ) state. For each candidate transaction, one validator is randomly selected as the verifier, while measurement angles are generated according to the protocol constraint that their sum is an integer multiple of Π . Each validator performs the corresponding quantum measurement and returns a binary measurement outcome. Instead of using conventional majority voting or independent approval from validators, the verifier determines consensus from the XOR parity of the collected measurement outcomes and compares it with the parity prescribed by the measurement angle condition. The implemented system combines this quantum validation stage with classical blockchain operations, including SHA256 transaction hashing, block generation, validator records, and replication of accepted blocks across a shared ledger. The transaction hash is retained for classical data integrity, while quantum measurement outcomes are used specifically for the consensus decision. The quantum execution layer is separated from the blockchain application through a backend interface, enabling the protocol to be executed using a Qiskit based quantum simulator. The resulting system provides a practical proof of concept for reproducing and integrating the quantum validation procedure described in prior work within an executable blockchain environment. It also establishes a software framework that can be extended for experimental evaluation and eventual execution on quantum hardware. For 20

independent simulations trials the quantum validation achieved a 100% parity since it was in an ideal environment, and the average execution time was $3.25 \pm 0.05S$ per trial.

2 Introduction

Blockchain technology ensures distributed state preservation, in which each of several players contributes to maintaining a common ledger without relying on a single governing authority. In a classical blockchain, there is a combination of different components that act together to achieve coordination. These components include transaction representation, cryptographic hashing, authentication, communication among the participants' nodes, transaction confirmation, achieving consensus, transitioning the state, and maintaining the distributed ledger. When a correct transaction is confirmed, it is put into the block according to the consensus and validity rules, and then the state of the blockchain is updated and preserved by the participants. Thus, the blockchain's reliability is a product of the interactions among different elements of the distributed system rather than the effectiveness of a particular cryptography method.

The principles of blockchain originate from previously existing ideas in area of cryptography and decentralized digital systems. In 1991, researchers Haber and Stornetta presented a solution for cryptographically linking timestamps of digital documents which became the paramount idea of tamper proof series of records [1]. Then there were a number of ventures such as b-money and Bit Gold which were aimed at exploring decentralized digital values and means of recording digital processes without needing the existence of a central authority. The concepts elaborated in the course of these developments later proved to become the building blocks of blockchain techniques and included cryptographic integrity, decentralized activities, distributing records and shared states.

A key change emerged with the release of Satoshi Nakamoto's Bitcoin: A Peer-to-Peer Electronic Cash System [2] and the start of the Bitcoin system in 2009. The introduction of Bitcoin included features like the linking of a hash, digital signatures, peer-to-peer technology, transaction validation, and proof of work making Bitcoin a working decentralized ledger system. Hence, the importance of this development exceeds the limits of the chained blocks allowing us to see how transaction integrity, communication, consensus, and replicated state can work without a central administrator. After that, blockchain technology developed into something more than just money transfer systems. Ethereum developed in 2013-2014 and was released in 2015, expanded that system by adding a function of smart contracts making blockchain not only a method of transferring money but rather a platform for decentralized applications.

The classical blockchain framework creates the basis upon which quantum blockchain studies have evolved. Specifically, aspects like transaction validity, participation of validators, consensus, state transformation, and maintaining the ledger still apply, even when quantum technology is involved. Nonetheless, quantum computing has added a new feature to the problem of blockchain studies, since quantum computing changes the assumptions regarding the security of classic blockchains and also allows using quantum data as a resource

in blockchain systems.

Quantum computing relies on the manipulation of quantum states which have characteristics like superposition, interference, and entanglement. The creation of quantum algorithms has shown that quantum computers could offer computational benefits for certain types of problems. The algorithm proposed by Shor in 1994 proved that integer factorization and discrete logarithms could in principle be solved efficiently on a quantum computer that is sufficiently capable [3]. As these problems form the basis of important public-key cryptographic systems, the finding indicated a possible long-term danger to digital signatures and to other cryptographic methods used in blockchain systems. Grover's algorithm, which was introduced in 1996, showed a quadratic speedup for unstructured search, the complexity being approximately $O(\sqrt{N})$ as opposed to $O(N)$ for a simple classical search [4]. While the implications of Grover's algorithm are different from those of Shor's, it is still relevant to blockchain since hashing and computational search are part of several blockchain security and consensus mechanisms. As a consequence, the tie between quantum computing and blockchain took form along two main lines described above. One direction is defensive and aims at ensuring the safety of existing classical blockchain solutions from future quantum attacks. The other direction is constructive and studies the possibilities of working quantum computation, telecommunication, and quantum mechanics themselves in the realm of blockchain technology. The first direction led to studies of post-quantum blockchain systems and quantum-protected blockchain architecture, while the second direction gave rise to quantum-augmented and purely quantum blockchain technologies. Kiktenko et al. presented one of the first responses to quantum threat by maintaining the classical blockchain structure but changing some components to increase their security. Kiktenko et al. considered quantum-protected blockchain architecture containing Quantum Key Distribution together with an ordinary blockchain setup. The findings showed the feasibility of using Quantum Key Distribution via urban fiber networks, implying that quantum communication can ensure security without the need to turn transactions, blocks, or records into quantum states [5].

Another different course is post-quantum cryptography (PQC) wherein cryptographic primitives that are exposed to quantum attacks have been substituted by those that can withstand the quantum attacks that gave birth to the QuantumShield BC which incorporates quantum random number generation technology, quantum key distribution technology and quantum Byzantine Fault Tolerance [6] as well as utilizing Dual Layer Signature Mechanism [7]. Hence, both of their approaches aim to tackle the quantum threat while preserving a basic classical blockchain system. It is essential to note that there is a difference: quantum-secured blockchain uses quantum technology applied to one or more of the relevant security functions whereas post-quantum blockchain changes the cryptographic premises of classical technology regardless of quantum technologies employed.

This development brings us to an important question: instead of using a quantum technology purely for the protection of a classical blockchain, can quantum information itself come into use for representation, validation, or becoming a part of the consensus mechanism of the blockchain? This question represents the emergence of fundamentally quantum or quantum-native blockchain architectures, where quantum state, quantum correlations,

phase information, or entanglement will be used to represent information, validate blocks, or decide whether a transition should be accepted.

As a result, there has been considerable growth in various threads of research based on quantum blockchain technology. One such research area deals with the concept of quantum block design, with an emphasis on mapPing classical blockchain data onto quantum states. A recent study of Banerjee, Mukherjee, and Panigrahi focused on quantum blockchain technology that is based on quantum hypergraphs and which incorporates phase-based methods for quantum representation of blockchain information. Their research has raised a number of significant questions on the questions of transforming classical data into quantum states and establishing relationships between the subsequent states [8].

A different aspect is instead oriented to achieving quantum consensus and validation based on entanglement. In ordinary blockchains, the nodes that participate in the blockchain can apply their own validation to a transaction and thus validate it through the vote or any form of agreement. The entanglement method is about the question of whether from the correlations existing among the measurements done by quantum systems, evidence can be obtained confirming that the participants have a legitimate quantum state. In this respect, GHZ states are very important because they provide for the correlations involving different quantum systems. Each qubit can be associated with a participating node, and their results can be studied to see whether the measured correlations correlate with the expected one.

Consequently, this indicates a significant shift in the validator's function. In a classical validation process, Transaction→Individual Validation→Vote/Approval, while an entanglement based procedure can be shown as Shared Quantum State→Distributed Measurement→Correlated Outcomes→Quantum Validation Condition. Thus, the validator can be thought of as not being an independent voter. The validator becomes a participant in a quantum measurement process with the final decision based on the quality of information between measurement outcomes, which is one of the key points of this work. In addition to these quantum-native method, hybrid quantum-classical architecture is being actively developed since the current state of quantum hardware and quantum networking cannot perform all required functions of a classical distributed ledger. In hybrid architecture, classical systems can do everything from managing transaction data, hashing, metadata, networking, application logic, and persistent ledger storage, while quantum resources involved for separate functions like state preparation, entanglement generation, validation, authentication, and signatures. This means that "quantum blockchain" is not a term for a well-defined architecture but may refer to various replacements of the conventional blockchain architecture or its components with quantum technology. In other words, it can be introduced in order to ensure protection against quantum attacks, transmission of quantum messages, creation of quantum data, verification of the states of the system including that it has achieved consensus, etc. Thus, putting a quantum technology into a blockchain system is not a sufficient condition for saying that this system is quantum.

During this development, the work of Del Rajan and Matt Visser, titled Quantum Blockchain Using Entanglement in Time,[9] serves as the basis for this study. The work was first uploaded to arXiv in 2018 and then published in Quantum Reports in 2019. The main

issue is that quantum entanglement is applied as one of the building blocks of blockchain but not just as a means of increasing security. The suggested construction is based on the concept of a temporal Greenberger–Horne–Zeilinger (GHZ) state that builds correlations among quantum systems that are related in time and incorporates those correlations into the blockchain structure.

The Rajan-Visser framework is important because it establishes a link between quantum entanglement and the process of blockchain validation. In accordance with the framework, an individual among the nodes engaged in the process is assigned the role of a verifier, and the quantum block is distributed to all the participating nodes. The verifier presents measurement parameters, following the premise of the protocol, and each of the other participating nodes executes the measurement on its own quantum instance. Afterwards, the nodes send a binary result of their measurements to the verifier, and the verifier processes their results according to the XOR/parity condition based on the measurement angles.

The importance of this mechanism rests on three interconnected components: Shared Quantum State→Distributed Measurements→Collective Validation Condition. The shared quantum state creates correlations among the participating network nodes, the distributed measurements provide a method of turning quantum signals into classical information, and the parity condition translates the correlated classical information into a blockchain-level validation.

Simultaneously, the work done by Rajan–Visser provides a conceptual framework for validation rather than an actual application of a traditional blockchain network. The authors indicate the need of development of low-level consensus and quantum network implementation details. This distinction creates a crucial implementation-level background for further work: the conceptual mechanism explains how a quantum correlation can be associated with validation while the engineering problem is how to connect this mechanism to transaction processing, validator’s management, block generation as well as ledger’s status and the real quantum execution.

The current research investigates this implementation and integration layer. Instead of introducing a new entanglement-based consensus method, this research creates an executing hybrid environment of blockchain that is based on the quantum verification process as described in the Rajan-Visser framework. This serves as a foundation for creating an executable workflow of transition from quantum and classic verification processes. The contribution made in the present study can therefore be regarded as an implementation study of a classical quantum verification technique. The proof-of-concept thus created forms grounds for future examinations with regard to quantum noise, multiple measurements, dishonest or defective participants, different validator configurations, scalability, and depositing on a real quantum machine.

3 Methods

A hybrid classical-quantum blockchain prototype was built here. Conventional blockchain tasks like transaction handling, hashing, block assembly, and ledger copying stay unchanged,

but validation uses an entanglement-based quantum method. This relies on the Theta-protocol from quantum blockchain studies, especially the GHZ-state check by Rajan and Visser. Six validators act as six people sharing a six-qubit GHZ state. For every new transaction, one validator Picks randomly to verify it, and six measurement angles follow the needed angle rule. One shared GHZ circuit forms, measurements hit all six qubits, giving one bit result per validator. These six bits go back to the blockchain part, where the checker tests the protocol's XOR parity rule. Only if that rule passes does the transaction move forward to make a block and lock into the shared ledger.

We retain the classical blockchain mechanisms wherever necessary. For example, we use classical mechanisms such as SHA-256 processing on each transaction to generate a deterministic “transaction hash” that supports classical data integrity/transaction identification mechanisms. Our use of the quantum mechanism is limited to the validation (or consensus) stage; we do not replace classical hashing with our quantum approach. In particular, the numeric representations derived from the SHA-256 transaction hash differ significantly from the angle measurements generated via the qubit's behavior under the Θ -protocol. In other words, we treat the transaction hash as part of the classical blockchain, while ensuring quantum measurement angles can be generated independent of the transaction hash value. These quantum measurement angles are created based upon the requirements for successful operation of the quantum validation process itself. Thus, we preserve the ability to separately evaluate performance aspects of the classical and quantum components.

At a high level, the classical layer handles aspects such as transaction processing, hashing, block building, and ledger management. Quantum components such as GHZ-state preparation, distributed measurements, correlated outcomes, and their corresponding validation condition are handled by the second layer of this structure. Finally, the actual execution of the quantum component itself — including all the elements of the computation discussed above — is decoupled from the rest of the blockchain application via a dedicated backend interface. This allows one (for example) to use a Qiskit-based quantum simulator that will run the circuit on behalf of the system and provide results (i.e., the measurement outcomes) back to the blockchain layer over an API connection. This allows a modular architecture where subsequent blockchain-level workflows could potentially plug into other quantum backends without having to redo all of the ledger implementation. In this case, the implementation is a computational proof of concept instead of an actual realization of a quantum network. The paper on which the project is based explains the quantum blockchain as a conceptual design that uses GHZ-state verification as a method of block verification.

3.1 Experimental Environment

Computing Platform: The goal of the experiment is to assess the blockchain process and the quantum verification procedure without relying on cloud quantum technology. Therefore, the actual running of the circuit is done locally with the help of Qiskit Aer. In that way, the experimental environment consists of two layers cooperating with each other: Classical Application Layer + Local Quantum Execution Layer In the classical layer, the transactions, validators, blocks, consensus state, and ledger synchronization are controlled, while the quantum layer develops and runs the six-qubit verification circuit and returns the six measurement data for block acceptance assessment.

Software environment: The principal software components used for the quantum implementation are given in Table 1.

Table 1: Principal components used

Component	Version / Configuration	Role
Python	Python environment used for the Qiskit backend	Quantum execution
Qiskit	2.2.3	Quantum circuit construction
Qiskit Aer	0.17.2	Local quantum simulation
NumPy	2.4.6	Numerical operations
SciPy	1.17.1	Scientific-computing environment
HTML5	–	User-interface layer
CSS	–	User-interface presentation
JavaScript	–	Blockchain application and control logic

Quantum execution code specifically imports QuantumCircuit from Qiskit and AerSimulator from Qiskit Aer. The simulator backend accepts six measurement angles, constructs a six-qubit circuit, executes it in one shot, and returns six binary measurement results. Use of local Aer backend is intentional. No IBM Quantum hardware was utilized in the experiment and no cloud execution was needed for the consensus Pipeline. Thus, hardware queue time, cloud-access requirements, and hardware-related noise were removed from the baseline version.

3.2 System Architecture

The entire system architecture is comprised of three major layers which is described in the Figure 1. The figure also gives an expanded view of the overall workflow that happens within the 3 layers.

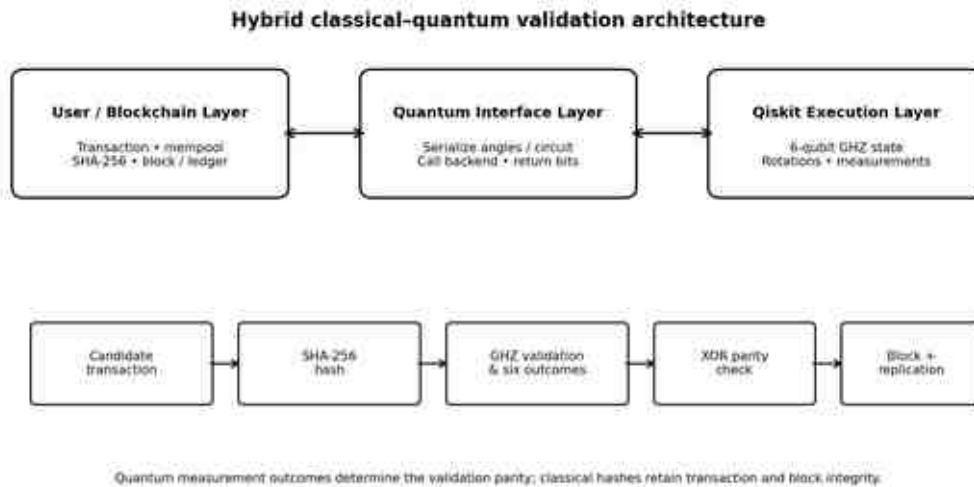


Figure 1: 3 Layer Architecture

The user and blockchain layer consists of the transaction model, mempool, validator nodes, blocks, public ledger, conventional hash functions, and consensus mechanism. The quantum interface layer is responsible for transformation of the blockchain-based measurement parameters to the serialized six-qubit circuit description and invokes the quantum backend. The Qiskit execution layer forms the actual QuantumCircuit, prepares the GHZ state, performs local angle-dependent measurements, runs the circuit using AerSimulator and obtains six measurement results. This separation is reflected in the architecture of the software.

3.3 Blockchain Network Configuration

As illustrated in Figure 2, our blockchain has one primary user node with six validator nodes (denoted by $V = V1, V2, V3, V4, V5, V6$). These six validators have their unique identifiers, network addresses, states and their own coPies of the blockchain ledger. We map these six validator nodes to the six qubits in our quantum verification circuit where $V_i \leftrightarrow q_i$ for $i \in 1, \dots, 6$. Therefore, there are six validators $\equiv$ six quantum participants $\equiv$ six qubits. To emphasize this, note that all the validators are not individual quantum circuits but only part of one large shared quantum circuit – thus, having six quantum participants means they share one big quantum circuit representing the collective verification state. Also, each quantum participant (validator) can give us one outcome of the measurements made on them (Y_i).



Figure 2: Blockchain Network Arrangement

Our block chain has various different state structures including Shared Ledger, Mempool and Local Validator Ledgers. The Shared Ledger shows up as an authoritative source of truth held within the application itself. While every validator holds their own synchronized copy of the blockchain's ledger. In order to avoid concurrent processes of more than one transaction we implemented a lock around the process which allows us to process just the very first transaction in the Mempool before moving on to the second.

3.4 Genesis Block Initialization

Prior to starting normal transaction processing, the blockchain needs to be initialized via a genesis block. A genesis transaction gets generated internally, which sets an initial balance to the primary user. Its SHA-256 transaction hash gets computed. Its status gets marked as committed, and based on that, its corresponding genesis block gets created having zero-value for its previous-hash field. This genesis block will get inserted in the shared ledger and simultaneously replicated across the local ledgers of all validators. Thus, this genesis block is primarily utilized to initialize/establish the initial common states of all six- validator networks. The protocol is such that $B_0 \rightarrow \{L_1, L_2, L_3, L_4, L_5, L_6\}$. (In other words, the quantum Θ -protocol isn't used for creating this initial bootstraping state - the quantum validation mechanism will be applied to subsequently submitted candidates.) By having this property, we prevent our initialization phase from requiring an invalid consensus state. To ensure this happens correctly, our implementation explicitly synchronized the genesis block with all 6 validators prior to allowing normal transaction submission.

3.5 Transaction flow

Transaction initiation begins from the perspective of the primary user who submits the transaction containing relevant parameters such as sender, recipient, amount to send, transaction fee, timestamp, metadata, and transaction identifier. Prior to insertion into our validation Pipeline, basic classical checks are run ensuring positivity of the amount parameter and that the sender has sufficient balance to pay for both the amount and transaction fees (in

order). Once validated, this transaction will be placed into the mempool which serves as a temporary storage system for transactions which have been created but not yet committed to the blockchain's stateful ledger. For simplicity reasons during our experiments, we've designed the Pipeline to only handle processing of one transaction at a time as shown in Eq 1,

$$T_0 \rightarrow \text{Mempool} \xrightarrow[\text{Classical}]{\text{Quantum}} \text{Block } T_0 \quad (1)$$

Such an approach ensures that we may cleanly associate all results directly attributable to the quantum validation with just the one candidate transaction. Prior to running quantum validation on a transaction, it was required to compute its hash in accordance with SHA-256 producing a 256-bit digest representing the contents of the transaction. In addition to the hexadecimal encoding provided natively by the algorithm, we've recorded the same result using the binary and decimal bases. To avoid losing any bit due to JavaScript rounding issues when storing large integers, the latter is computed via BigInt instead of relying upon the native Number class. It should also be noted that when considering the binary format of the hash, care is taken to maintain the exact 256 bits of output including leading zeros. Finally, we derive a classical transaction associated angle defined as:

$$\Theta_{tx} = \frac{D}{2^{256} - 1} * \frac{\pi}{4}, 0 \leq \Theta_{tx} \leq \frac{\pi}{4} \quad (2)$$

From Eq 2, D represents the decimal equivalent value of the given 256-bit transaction hash. This guarantees the resultant angle falls within $[0, \pi/4]$. It should be noted that the value above is never considered part of the set of quantum measurement angles Θ_j participating in the consensus protocol. This distinction ensures that the quantum protocol remains separate from the classical tx.hash representation. Consequently, the transaction-associated parameter differs from the measurement angles, i.e., $\Theta_{tx} \neq \Theta_j$.

Once the transaction hash has been generated, one of the six validators is randomly selected to act as the verifier for the current validation round. Let the randomly selected validator index be defined as:

$$k = \lfloor 6r \rfloor, \quad r \sim \mathcal{U}(0, 1) \quad (3)$$

Thus, in Eq 3, $k \in \{0, 1, \dots, 5\}$, and validator V_k acts as the verifier for the current transaction. It should also be noted that the remaining validators are not merely independent classical voters; rather, each validator acts as a quantum node associated with its respective qubit of the shared GHZ state.

3.6 Generation of the Six Quantum Measurement Angles

In this case we need six measurement angles. Implementing this configuration you will generate 5 random positive values which are going to be used to define the measure angles, making sure these five angles cover an approximate 95% of the interval associated with π . For the last one we have calculated its value considering the rest of the range.

Let $r_1, r_2, \dots, r_5$ be five random values generated by the implementation, and let

$$R = \sum_{i=1}^5 r_i, \quad \alpha = \frac{0.95}{R} \quad (4)$$

In Eq 4 we have the sum of the 5 random values and a scaling coefficient factor is defined by α .

The first five measurement angles are calculated by Eq 5

$$\Theta_i = \Pi * \alpha * r_i, \quad i = 1, \dots, 5. \quad (5)$$

The final angle is calculated by the Eq 6

$$\Theta_6 = \Pi - \sum_{i=1}^5 \Theta_i, \quad \sum_{i=1}^6 \Theta_i = \Pi \quad (6)$$

We have implemented up to the normal floating point representation. To enforce the six angle constraint we have made this restriction explicit. I.e., the function will reject configurations containing anything but 6 validators. Given that the sum of the six measurement angles is constrained to equal Π , the right-hand side of the Θ -protocol parity condition becomes 1 as defined by Eq 7

$$\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \bmod 2 = \frac{\Pi}{\Pi} \bmod 2 = 1 \quad (7)$$

Therefore, we expect the six-validator experiment to produce an output satisfying the following property:

$$\left(\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \right) \bmod 2 = 1 \quad (8)$$

This rather strange-looking design of Eq 8 leads to an extremely simple final consensus condition of just requiring that the XOR of these six measured binary outcomes equals 1.

3.7 Construction of the Six-Qubit GHZ State

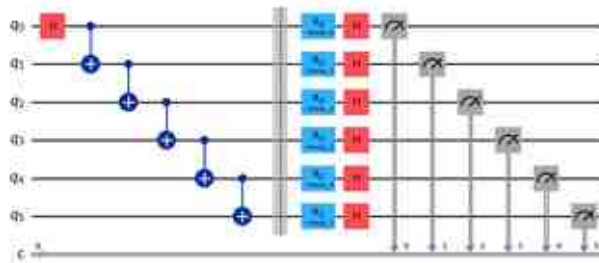


Figure 3: Qiskit Implementation of 6 qubit GHZ circuit

As in the Figure 3, Quantum validation circuit has six qubits and six classical measurement bits $(q_0, q_1 \dots, q_5)$ and $(C_0, c_1 \dots, c_5)$. Initially these are set to $|000000\rangle$.

Then applying a Hadamard gate on the qubit q_0 we get Eq 9,

$$|000000\rangle \rightarrow \frac{1}{\sqrt{2}}|000000\rangle + |100000\rangle \quad (9)$$

Next applying a sequence of linear CNOT entanglement chain of gates yields the desired six qubit GHZ state which gives Eq 10,

$$|GHZ_6\rangle = \frac{1}{\sqrt{2}}(|000000\rangle + |111111\rangle) \quad (10)$$

Instead of implementing six independent circuits for each participant as would be expected from an ordinary GHZ circuit, it's implemented here as a single quantum circuit that creates a commonly shared quantum state between six validators.

3.8 Final Integration into the classical flow

The quantum-validation stage performs measurements in the Θ basis by applying an $R_z(-\Theta_j)$ operation followed by a Hadamard gate H to each validator qubit, yielding a binary measurement result $Y_j \in 0, 1$.

Once computed, the browser application transmits the six measurement angles and the circuit description to the Python-based quantum backend over the network connection. After processing these inputs, the quantum backend returns the resulting six-bit vector as defined in Eq 11, whose result can be mapped to the corresponding validators.

$$\mathbf{Y} = [Y_1, Y_2, Y_3, Y_4, Y_5, Y_6], \quad V_i \leftrightarrow (q_i, \Theta_i, Y_i) \quad (11)$$

These measured bits represent a single quantum-validation event rather than independent approvals. Accordingly, consensus is calculated using the measured parity by Eq 12, where $\oplus$ denotes the exclusive OR (XOR) operation.

$$P_{\text{measured}} = Y_1 \oplus Y_2 \oplus Y_3 \oplus Y_4 \oplus Y_5 \oplus Y_6 \quad (12)$$

The measured parity is compared against the protocol condition and measurement angle constraint which gives $P_{\text{expected}} = 1$ by Eq 13.

$$P_{\text{expected}} = \left(\frac{1}{\Pi} \sum_{j=1}^6 \Theta_j \right) \bmod 2, \quad \sum_{j=1}^6 \Theta_j = \Pi \quad (13)$$

Therefore, for the transaction to transition from the **PENDING** state to the **VALIDATED** state, the measured parity must satisfy the Eq 14.

$$P_{\text{measured}} = P_{\text{expected}} = 1 \quad (14)$$

Finally, after storing the quantum-verification results along with the transaction data itself within our system, the next step involves generating a classical block consisting of the current transaction, its associated previous block hash, and transaction hash, as well as both

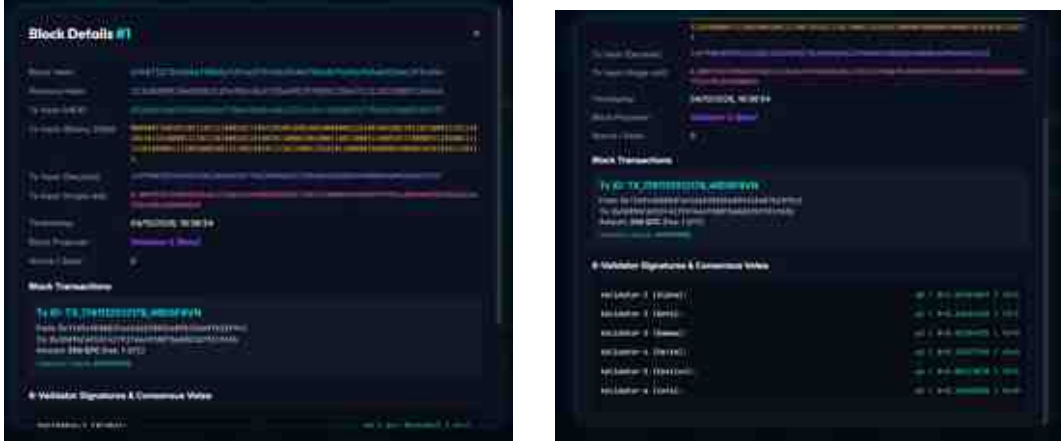
the proposer and the records of verification provided by each validator. This last block can subsequently be secured using SHA-256 hashing. Importantly, though, one should note that the role of proposer comes from deterministic selection among those participating as validators; importantly, this position differs from that randomly chosen verifier who participated during the quantum validation/verification process. With this final step completed, the proposed block becomes part of the shared ledger and is distributed to all participants in the six-validator network. Ledger updates propagate across the distributed system by ensuring that each participant receives an identical copy of the validated block before appending it to their respective local ledger.

Following block propagation, user's balances are updated accordingly, and the contents of the transaction mempool are synchronized with the latest ledger state. To maintain system integrity and ensure correct operation, additional validation checks verify the relationships among the previous block hash, the current block hash, and the transaction hashes. These checks help ensure consistency across the distributed ledger and detect discrepancies between local ledger copies.

4 Results



Figure 4: Shared Ledger



(a) Block information

(b) Block Validator Information

Figure 5: Newly added block details

The prototype successfully demonstrates an end to end transaction validation and block commitment cycle over the 6-node validator network after syncing up with the genesis block. As shown in the figure, when the submitted transaction has been validated with the blockchain protocol, the transaction's SHA-256 hash results into a hash prefix and decimal representation. In this case, we get 0.0097535759 rad as our computed transaction angle. Quantum validation is performed on the Delta node, where a random selection process designates the verifier for the current validation round. Subsequently, six measurement angles are generated such that their combined sum satisfies the required constraint $\sum_{i=1}^6 \Theta_i = \Pi$ radians. The six validator nodes produce the measurement vector as in Eq 15

$$\mathbf{Y} = [1, 1, 0, 0, 1, 0] \quad (15)$$

The measured XOR parity is calculated as

$$P_{\text{measured}} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1 \quad (16)$$

Since the measured parity in Eq16 matches the expected parity, $P_{\text{expected}} = 1$, the Θ -protocol consensus condition is satisfied, allowing the transaction to be accepted for block construction, as illustrated in Figure 5.

Block 1 is subsequently constructed by the Beta node and linked to the genesis block through its previous-block hash reference. Following synchronization across the six validator ledgers, each ledger incorporates the new block, increasing its height from 1 to 2, as illustrated in Figure 4. This was repeated for another 19 times and all the 20 independent simulations trials of the quantum validation achieved a 100% parity since it was in an ideal environment, and the average execution time was $3.25 \pm 0.05S$ per trial.

5 Discussion

The current work is simulator based and does not demonstrate practical security or real-world quantum consensus. While demonstrating successful performance by GHZ based Validation using the above mentioned protocol we found out some constraints which are very crucial from the point of view of actual implementations. First, we see that the Verification condition (collective sum) requires the measurements angles should satisfy certain multiple of π i.e., 0 . An ideal simulator could ensure this constraint. However, real world Quantum Hardware suffers from finite precision control gates and cumulative angle errors leading to possible deviation in the effective sum j thus reducing the reliability of desired parity output. Second, even though the proposed framework provides us with an encoding mechanism for representing the blockchain record as a entangle quantum state, the proposed Θ -protocol only verify the GHZ correlation of the candidate state, instead of verifying the presence of the intended block information in the state. As per the paper's discussion they have described about how their classical block records were encoded into temporal Bell and GHZ States, but the part where you test if your supplied candidate State has the GHZ Correlations required is the Verification Stage here. So basically they showed off their Quantum Validation Mechanism. However, to build up a full-fledged Block Authentication Mechanism, we need to explicitly bind the classical block information we want to authenticate with the Candidate State to be Verified. This becomes really important because someone could simply go onto the Classical Blockchain Network, and tamper the block contents while keeping the underlying Quantum Consensus untouched. And since the Quantum Consensus doesn't care about what happened on the classical side of things, it just happily accepts the tampered block. Now the recent paper regarding Time Entanglements in 2026 [10] manages to overcome this binding challenge, however it runs into serious challenges as the Number Of Blocks Increase, hence working out ways to Combine both methodologies may form an open research problem in our near future scope of Research.

References

- [1] S. Haber and W. S. Stornetta, "How to time-stamp a digital document," *Journal of Cryptology*, vol. 3, pp. 99–111, 1991.
- [2] W. Zhao, "Blockchain technology: Development and prospects," *National Science Review*, vol. 6, pp. 369–373, 2019.
- [3] P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*. IEEE Computer Society Press, 2002.
- [4] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing (STOC '96)*. New York, NY, USA: ACM Press, 1996, pp. 212–219.
- [5] E. O. Kiktenko, N. O. Pozhar, M. N. Anufriev, A. S. Trushechkin, R. R. Yunusov, Y. V. Kurochkin, A. I. Lvovsky, and A. K. Fedorov, "Quantum-secured blockchain," *Quantum Science and Technology*, vol. 3, p. 035004, 2018.

- [6] N. R. Reddy, S. Suryadevara, K. G. R. Reddy, R. Umamaheswari, R. Guttula, and R. Kotoju, "Quantum secured blockchain framework for enhancing post quantum data security," *Scientific Reports*, vol. 15, p. 31048, 2025.
- [7] K. W. Goh, B. U. I. Khan, A. R. Khan, D. S. Putra, F. R. P. P., and M. A. Bhuyian, "Eliminating single points of trust: A hybrid quantum and post-quantum blockchain with distributed key generation," *Scientific Reports*, vol. 16, p. 1286, 2026.
- [8] S. Banerjee, A. Mukherjee, and P. K. Panigrahi, "Quantum blockchain using weighted hypergraph states," *Physical Review Research*, vol. 2, no. 1, 2020.
- [9] D. Rajan and M. Visser, "Quantum blockchain using entanglement in time," *Quantum Reports*, vol. 1, pp. 3–11, 2019.
- [10] R. Konara, K. D. Zoysa, A. Mahasinghe, A. Sayakkara, and N. Ranasinghe, "Time-entangled quantum blockchain with phase encoding for classical data," *Quantum Reports*, vol. 8, p. 69, 2026.

Test Author

Journal Final Draft.pdf



Document Details

Submission Date

Oct 4, 2026, 6:48 PM GMT

Download Date

Oct 4, 2026, 6:50 PM GMT

File Name

Journal Final Draft.pdf

File Size

297.0 KB

5,803 Words

32,682 Characters

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

-  **54 Not Cited or Quoted 5%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 4%  Internet sources
- 2%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **54 Not Cited or Quoted 5%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 4%  Internet sources
- 2%  Publications
- 0%  Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	cora.ucc.ie	1%
2	Internet	researchspace.auckland.ac.nz	1%
3	Publication	Deterministic multi-qubit entanglement in a quantum network	1%
4	Internet	www.psgo.gov.hk	1%
5	Publication	Time-Entangled Quantum Blockchain with Phase Encoding	1%
6	Internet	scholar.tecnico.ulisboa.pt	<1%
7	Internet	mro.massey.ac.nz	<1%
8	Publication	A review of quantum machine learning algorithms, applications, and emerging ...	<1%
9	Publication	BEAT: Blockchain-Enabled Accountable and Transparent Infrastructure Sharing ...	<1%
10	Publication	Trustworthy Foundations for Web3	<1%

11	Internet	books.google.com	<1%
12	Internet	edulearninghub.com	<1%
13	Internet	iris.unil.ch	<1%
14	Publication	The Quantum-Cryptographic Co-evolution	<1%
15	Publication	ExplainBench: A Benchmark Framework for Local Model	<1%
16	Internet	mdpi-res.com	<1%
17	Publication	Tax Policy Handbook for Crypto Assets	<1%
18	Publication	(Accessed: 5 September 2026)	<1%
19	Internet	www.disi.unitn.it	<1%
20	Publication	Evolution of blockchain consensus algorithms: a review on the latest milesto ...	<1%
21	Internet	w.pauldowling.me	<1%
22	Internet	publications.lib.chalmers.se	<1%
23	Publication	mainstream media. AWARI, 5, 1–12	<1%
24	Publication	Excessive Performance Target and Performance of Marketing Staff in Turbulent ...	<1%
25	Internet	www.language-and-society.org	<1%

26	Publication	Blockchain in the Electronics Industry for Supply Chain Management: A Survey	<1%
27	Publication	doi: ISSN 0350-137X. Archived from the original on 16 April	<1%
28	Internet	search.proquest.com	<1%
29	Internet	uxpod.com	<1%
30	Internet	backend.orbit.dtu.dk	<1%
31	Internet	www.enpress-publisher.com	<1%
32	Internet	eltchoutari.com	<1%
33	Internet	link.aps.org	<1%
34	Internet	eprint.iacr.org	<1%
35	Internet	csrc.nist.gov	<1%
36	Internet	www.jmlr.org	<1%

11

23

3

31

independent simulations trials the quantum validation achieved a 100% parity since it was in an ideal environment, and the average execution time was $3.25 \pm 0.05S$ per trial.

2 Introduction

Blockchain technology ensures distributed state preservation, in which each of several players contributes to maintaining a common ledger without relying on a single governing authority. In a classical blockchain, there is a combination of different components that act together to achieve coordination. These components include transaction representation, cryptographic hashing, authentication, communication among the participants' nodes, transaction confirmation, achieving consensus, transitioning the state, and maintaining the distributed ledger. When a correct transaction is confirmed, it is put into the block according to the consensus and validity rules, and then the state of the blockchain is updated and preserved by the participants. Thus, the blockchain's reliability is a product of the interactions among different elements of the distributed system rather than the effectiveness of a particular cryptography method.

The principles of blockchain originate from previously existing ideas in area of cryptography and decentralized digital systems. In 1991, researchers Haber and Stornetta presented a solution for cryptographically linking timestamps of digital documents which became the paramount idea of tamper proof series of records [1]. Then there were a number of ventures such as b-money and Bit Gold which were aimed at exploring decentralized digital values and means of recording digital processes without needing the existence of a central authority. The concepts elaborated in the course of these developments later proved to become the building blocks of blockchain techniques and included cryptographic integrity, decentralized activities, distributing records and shared states.

A key change emerged with the release of Satoshi Nakamoto's Bitcoin: A Peer-to-Peer Electronic Cash System [2] and the start of the Bitcoin system in 2009. The introduction of Bitcoin included features like the linking of a hash, digital signatures, peer-to-peer technology, transaction validation, and proof of work making Bitcoin a working decentralized ledger system. Hence, the importance of this development exceeds the limits of the chained blocks allowing us to see how transaction integrity, communication, consensus, and replicated state can work without a central administrator. After that, blockchain technology developed into something more than just money transfer systems. Ethereum developed in 2013-2014 and was released in 2015, expanded that system by adding a function of smart contracts making blockchain not only a method of transferring money but rather a platform for decentralized applications.

The classical blockchain framework creates the basis upon which quantum blockchain studies have evolved. Specifically, aspects like transaction validity, participation of validators, consensus, state transformation, and maintaining the ledger still apply, even when quantum technology is involved. Nonetheless, quantum computing has added a new feature to the problem of blockchain studies, since quantum computing changes the assumptions regarding the security of classic blockchains and also allows using quantum data as a resource

in blockchain systems.

Quantum computing relies on the manipulation of quantum states which have characteristics like superposition, interference, and entanglement. The creation of quantum algorithms has shown that quantum computers could offer computational benefits for certain types of problems. The algorithm proposed by Shor in 1994 proved that integer factorization and discrete logarithms could in principle be solved efficiently on a quantum computer that is sufficiently capable [3]. As these problems form the basis of important public-key cryptographic systems, the finding indicated a possible long-term danger to digital signatures and to other cryptographic methods used in blockchain systems. Grover's algorithm, which was introduced in 1996, showed a quadratic speedup for unstructured search, the complexity being approximately $O(\sqrt{N})$ as opposed to $O(N)$ for a simple classical search [4]. While the implications of Grover's algorithm are different from those of Shor's, it is still relevant to blockchain since hashing and computational search are part of several blockchain security and consensus mechanisms. As a consequence, the tie between quantum computing and blockchain took form along two main lines described above. One direction is defensive and aims at ensuring the safety of existing classical blockchain solutions from future quantum attacks. The other direction is constructive and studies the possibilities of working quantum computation, telecommunication, and quantum mechanics themselves in the realm of blockchain technology. The first direction led to studies of post-quantum blockchain systems and quantum-protected blockchain architecture, while the second direction gave rise to quantum-augmented and purely quantum blockchain technologies. Kiktenko et al. presented one of the first responses to quantum threat by maintaining the classical blockchain structure but changing some components to increase their security. Kiktenko et al. considered quantum-protected blockchain architecture containing Quantum Key Distribution together with an ordinary blockchain setup. The findings showed the feasibility of using Quantum Key Distribution via urban fiber networks, implying that quantum communication can ensure security without the need to turn transactions, blocks, or records into quantum states [5].

Another different course is post-quantum cryptography (PQC) wherein cryptographic primitives that are exposed to quantum attacks have been substituted by those that can withstand the quantum attacks that gave birth to the QuantumShield BC which incorporates quantum random number generation technology, quantum key distribution technology and quantum Byzantine Fault Tolerance [6] as well as utilizing Dual Layer Signature Mechanism [7]. Hence, both of their approaches aim to tackle the quantum threat while preserving a basic classical blockchain system. It is essential to note that there is a difference: quantum-secured blockchain uses quantum technology applied to one or more of the relevant security functions whereas post-quantum blockchain changes the cryptographic premises of classical technology regardless of quantum technologies employed.

This development brings us to an important question: instead of using a quantum technology purely for the protection of a classical blockchain, can quantum information itself come into use for representation, validation, or becoming a part of the consensus mechanism of the blockchain? This question represents the emergence of fundamentally quantum or quantum-native blockchain architectures, where quantum state, quantum correlations,

phase information, or entanglement will be used to represent information, validate blocks, or decide whether a transition should be accepted.

As a result, there has been considerable growth in various threads of research based on quantum blockchain technology. One such research area deals with the concept of quantum block design, with an emphasis on mapPing classical blockchain data onto quantum states. A recent study of Banerjee, Mukherjee, and Panigrahi focused on quantum blockchain technology that is based on quantum hypergraphs and which incorporates phase-based methods for quantum representation of blockchain information. Their research has raised a number of significant questions on the questions of transforming classical data into quantum states and establishing relationships between the subsequent states [8].

A different aspect is instead oriented to achieving quantum consensus and validation based on entanglement. In ordinary blockchains, the nodes that participate in the blockchain can apply their own validation to a transaction and thus validate it through the vote or any form of agreement. The entanglement method is about the question of whether from the correlations existing among the measurements done by quantum systems, evidence can be obtained confirming that the participants have a legitimate quantum state. In this respect, GHZ states are very important because they provide for the correlations involving different quantum systems. Each qubit can be associated with a participating node, and their results can be studied to see whether the measured correlations correlate with the expected one.

Consequently, this indicates a significant shift in the validator's function. In a classical validation process, Transaction→Individual Validation→Vote/Approval, while an entanglement based procedure can be shown as Shared Quantum State→Distributed Measurement→Correlated Outcomes→Quantum Validation Condition. Thus, the validator can be thought of as not being an independent voter. The validator becomes a participant in a quantum measurement process with the final decision based on the quality of information between measurement outcomes, which is one of the key points of this work. In addition to these quantum-native method, hybrid quantum-classical architecture is being actively developed since the current state of quantum hardware and quantum networking cannot perform all required functions of a classical distributed ledger. In hybrid architecture, classical systems can do everything from managing transaction data, hashing, metadata, networking, application logic, and persistent ledger storage, while quantum resources involved for separate functions like state preparation, entanglement generation, validation, authentication, and signatures. This means that "quantum blockchain" is not a term for a well-defined architecture but may refer to various replacements of the conventional blockchain architecture or its components with quantum technology. In other words, it can be introduced in order to ensure protection against quantum attacks, transmission of quantum messages, creation of quantum data, verification of the states of the system including that it has achieved consensus, etc. Thus, putting a quantum technology into a blockchain system is not a sufficient condition for saying that this system is quantum.

During this development, the work of Del Rajan and Matt Visser, titled Quantum Blockchain Using Entanglement in Time,[9] serves as the basis for this study. The work was first uploaded to arXiv in 2018 and then published in Quantum Reports in 2019. The main

issue is that quantum entanglement is applied as one of the building blocks of blockchain but not just as a means of increasing security. The suggested construction is based on the concept of a temporal Greenberger-Horne-Zeilinger (GHZ) state that builds correlations among quantum systems that are related in time and incorporates those correlations into the blockchain structure.

The Rajan-Visser framework is important because it establishes a link between quantum entanglement and the process of blockchain validation. In accordance with the framework, an individual among the nodes engaged in the process is assigned the role of a verifier, and the quantum block is distributed to all the participating nodes. The verifier presents measurement parameters, following the premise of the protocol, and each of the other participating nodes executes the measurement on its own quantum instance. Afterwards, the nodes send a binary result of their measurements to the verifier, and the verifier processes their results according to the XOR/parity condition based on the measurement angles.

The importance of this mechanism rests on three interconnected components: Shared Quantum State→Distributed Measurements→Collective Validation Condition. The shared quantum state creates correlations among the participating network nodes, the distributed measurements provide a method of turning quantum signals into classical information, and the parity condition translates the correlated classical information into a blockchain-level validation.

Simultaneously, the work done by Rajan-Visser provides a conceptual framework for validation rather than an actual application of a traditional blockchain network. The authors indicate the need of development of low-level consensus and quantum network implementation details. This distinction creates a crucial implementation-level background for further work: the conceptual mechanism explains how a quantum correlation can be associated with validation while the engineering problem is how to connect this mechanism to transaction processing, validator's management, block generation as well as ledger's status and the real quantum execution.

The current research investigates this implementation and integration layer. Instead of introducing a new entanglement-based consensus method, this research creates an executing hybrid environment of blockchain that is based on the quantum verification process as described in the Rajan-Visser framework. This serves as a foundation for creating an executable workflow of transition from quantum and classic verification processes. The contribution made in the present study can therefore be regarded as an implementation study of a classical quantum verification technique. The proof-of-concept thus created forms grounds for future examinations with regard to quantum noise, multiple measurements, dishonest or defective participants, different validator configurations, scalability, and depositing on a real quantum machine.

3 Methods

A hybrid classical-quantum blockchain prototype was built here. Conventional blockchain tasks like transaction handling, hashing, block assembly, and ledger copying stay unchanged,

but validation uses an entanglement-based quantum method. This relies on the Theta-protocol from quantum blockchain studies, especially the GHZ-state check by Rajan and Visser. Six validators act as six people sharing a six-qubit GHZ state. For every new transaction, one validator Picks randomly to verify it, and six measurement angles follow the needed angle rule. One shared GHZ circuit forms, measurements hit all six qubits, giving one bit result per validator. These six bits go back to the blockchain part, where the checker tests the protocol's XOR parity rule. Only if that rule passes does the transaction move forward to make a block and lock into the shared ledger.

We retain the classical blockchain mechanisms wherever necessary. For example, we use classical mechanisms such as SHA-256 processing on each transaction to generate a deterministic “transaction hash” that supports classical data integrity/transaction identification mechanisms. Our use of the quantum mechanism is limited to the validation (or consensus) stage; we do not replace classical hashing with our quantum approach. In particular, the numeric representations derived from the SHA-256 transaction hash differ significantly from the angle measurements generated via the qubit's behavior under the Θ -protocol. In other words, we treat the transaction hash as part of the classical blockchain, while ensuring quantum measurement angles can be generated independent of the transaction hash value. These quantum measurement angles are created based upon the requirements for successful operation of the quantum validation process itself. Thus, we preserve the ability to separately evaluate performance aspects of the classical and quantum components.

At a high level, the classical layer handles aspects such as transaction processing, hashing, block building, and ledger management. Quantum components such as GHZ-state preparation, distributed measurements, correlated outcomes, and their corresponding validation condition are handled by the second layer of this structure. Finally, the actual execution of the quantum component itself — including all the elements of the computation discussed above — is decoupled from the rest of the blockchain application via a dedicated backend interface. This allows one (for example) to use a Qiskit-based quantum simulator that will run the circuit on behalf of the system and provide results (i.e., the measurement outcomes) back to the blockchain layer over an API connection. This allows a modular architecture where subsequent blockchain-level workflows could potentially plug into other quantum backends without having to redo all of the ledger implementation. In this case, the implementation is a computational proof of concept instead of an actual realization of a quantum network. The paper on which the project is based explains the quantum blockchain as a conceptual design that uses GHZ-state verification as a method of block verification.

3.1 Experimental Environment

Computing Platform: The goal of the experiment is to assess the blockchain process and the quantum verification procedure without relying on cloud quantum technology. Therefore, the actual running of the circuit is done locally with the help of Qiskit Aer. In that way, the experimental environment consists of two layers cooperating with each other: Classical Application Layer + Local Quantum Execution Layer In the classical layer, the transactions, validators, blocks, consensus state, and ledger synchronization are controlled, while the quantum layer develops and runs the six-qubit verification circuit and returns the six measurement data for block acceptance assessment.

Software environment: The principal software components used for the quantum implementation are given in Table 1.

Table 1: Principal components used

Component	Version / Configuration	Role
Python	Python environment used for the Qiskit backend	Quantum execution
Qiskit	2.2.3	Quantum circuit construction
Qiskit Aer	0.17.2	Local quantum simulation
NumPy	2.4.6	Numerical operations
SciPy	1.17.1	Scientific-computing environment
HTML5	–	User-interface layer
CSS	–	User-interface presentation
JavaScript	–	Blockchain application and control logic

Quantum execution code specifically imports QuantumCircuit from Qiskit and AerSimulator from Qiskit Aer. The simulator backend accepts six measurement angles, constructs a six-qubit circuit, executes it in one shot, and returns six binary measurement results. Use of local Aer backend is intentional. No IBM Quantum hardware was utilized in the experiment and no cloud execution was needed for the consensus Pipeline. Thus, hardware queue time, cloud-access requirements, and hardware-related noise were removed from the baseline version.

3.2 System Architecture

The entire system architecture is comprised of three major layers which is described in the Figure 1. The figure also gives an expanded view of the overall workflow that happens within the 3 layers.

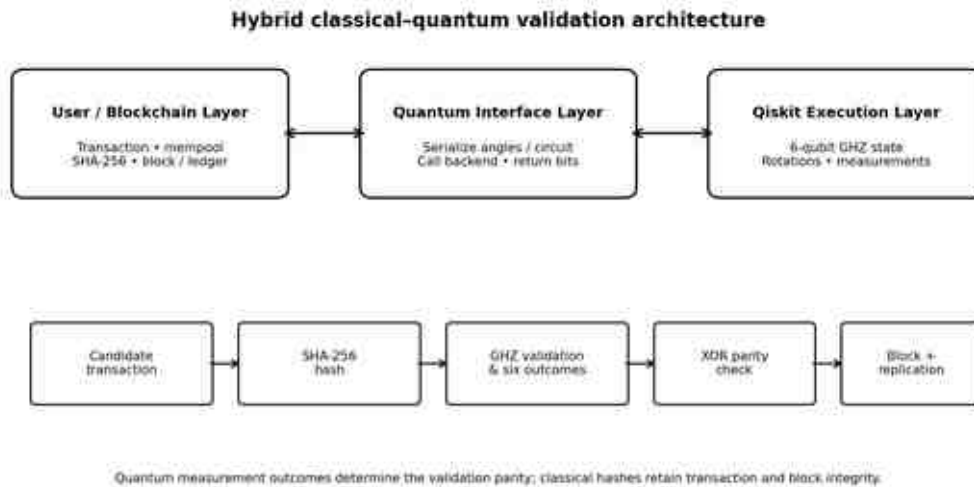


Figure 1: 3 Layer Architecture

The user and blockchain layer consists of the transaction model, mempool, validator nodes, blocks, public ledger, conventional hash functions, and consensus mechanism. The quantum interface layer is responsible for transformation of the blockchain-based measurement parameters to the serialized six-qubit circuit description and invokes the quantum backend. The Qiskit execution layer forms the actual QuantumCircuit, prepares the GHZ state, performs local angle-dependent measurements, runs the circuit using AerSimulator and obtains six measurement results. This separation is reflected in the architecture of the software.

3.3 Blockchain Network Configuration

As illustrated in Figure 2, our blockchain has one primary user node with six validator nodes (denoted by $V = V1, V2, V3, V4, V5, V6$). These six validators have their unique identifiers, network addresses, states and their own copies of the blockchain ledger. We map these six validator nodes to the six qubits in our quantum verification circuit where $V_i \leftrightarrow q_i$ for $i \in 1, \dots, 6$. Therefore, there are six validators $\equiv$ six quantum participants $\equiv$ six qubits. To emphasize this, note that all the validators are not individual quantum circuits but only part of one large shared quantum circuit – thus, having six quantum participants means they share one big quantum circuit representing the collective verification state. Also, each quantum participant (validator) can give us one outcome of the measurements made on them (Y_i).



Figure 2: Blockchain Network Arrangement

Our block chain has various different state structures including Shared Ledger, Mempool and Local Validator Ledgers. The Shared Ledger shows up as an authoritative source of truth held within the application itself. While every validator holds their own synchronized copy of the blockchain's ledger. In order to avoid concurrent processes of more than one transaction we implemented a lock around the process which allows us to process just the very first transaction in the Mempool before moving on to the second.

3.4 Genesis Block Initialization

Prior to starting normal transaction processing, the blockchain needs to be initialized via a genesis block. A genesis transaction gets generated internally, which sets an initial balance to the primary user. Its SHA-256 transaction hash gets computed. Its status gets marked as committed, and based on that, its corresponding genesis block gets created having zero-value for its previous-hash field. This genesis block will get inserted in the shared ledger and simultaneously replicated across the local ledgers of all validators. Thus, this genesis block is primarily utilized to initialize/establish the initial common states of all six- validator networks. The protocol is such that $B_0 \rightarrow \{L_1, L_2, L_3, L_4, L_5, L_6\}$. (In other words, the quantum Θ -protocol isn't used for creating this initial bootstraping state - the quantum validation mechanism will be applied to subsequently submitted candidates.) By having this property, we prevent our initialization phase from requiring an invalid consensus state. To ensure this happens correctly, our implementation explicitly synchronized the genesis block with all 6 validators prior to allowing normal transaction submission.

3.5 Transaction flow

Transaction initiation begins from the perspective of the primary user who submits the transaction containing relevant parameters such as sender, recipient, amount to send, transaction fee, timestamp, metadata, and transaction identifier. Prior to insertion into our validation Pipeline, basic classical checks are run ensuring positivity of the amount parameter and that the sender has sufficient balance to pay for both the amount and transaction fees (in

order). Once validated, this transaction will be placed into the mempool which serves as a temporary storage system for transactions which have been created but not yet committed to the blockchain's stateful ledger. For simplicity reasons during our experiments, we've designed the Pipeline to only handle processing of one transaction at a time as shown in Eq 1,

$$T_0 \rightarrow \text{Mempool} \xrightarrow[\text{Classical}]{\text{Quantum}} \text{Block } T_0 \quad (1)$$

Such an approach ensures that we may cleanly associate all results directly attributable to the quantum validation with just the one candidate transaction. Prior to running quantum validation on a transaction, it was required to compute its hash in accordance with SHA-256 producing a 256-bit digest representing the contents of the transaction. In addition to the hexadecimal encoding provided natively by the algorithm, we've recorded the same result using the binary and decimal bases. To avoid losing any bit due to JavaScript rounding issues when storing large integers, the latter is computed via BigInt instead of relying upon the native Number class. It should also be noted that when considering the binary format of the hash, care is taken to maintain the exact 256 bits of output including leading zeros. Finally, we derive a classical transaction associated angle defined as:

$$\Theta_{tx} = \frac{D}{2^{256} - 1} * \frac{\pi}{4}, 0 \leq \Theta_{tx} \leq \frac{\pi}{4} \quad (2)$$

From Eq 2, D represents the decimal equivalent value of the given 256-bit transaction hash. This guarantees the resultant angle falls within $[0, \pi/4]$. It should be noted that the value above is never considered part of the set of quantum measurement angles Θ_j participating in the consensus protocol. This distinction ensures that the quantum protocol remains separate from the classical tx.hash representation. Consequently, the transaction-associated parameter differs from the measurement angles, i.e., $\Theta_{tx} \neq \Theta_j$.

Once the transaction hash has been generated, one of the six validators is randomly selected to act as the verifier for the current validation round. Let the randomly selected validator index be defined as:

$$k = \lfloor 6r \rfloor, \quad r \sim \mathcal{U}(0, 1) \quad (3)$$

Thus, in Eq 3, $k \in \{0, 1, \dots, 5\}$, and validator V_k acts as the verifier for the current transaction. It should also be noted that the remaining validators are not merely independent classical voters; rather, each validator acts as a quantum node associated with its respective qubit of the shared GHZ state.

3.6 Generation of the Six Quantum Measurement Angles

In this case we need six measurement angles. Implementing this configuration you will generate 5 random positive values which are going to be used to define the measure angles, making sure these five angles cover an approximate 95% of the interval associated with π . For the last one we have calculated its value considering the rest of the range.

Let $r_1, r_2, \dots, r_5$ be five random values generated by the implementation, and let

$$R = \sum_{i=1}^5 r_i, \quad \alpha = \frac{0.95}{R} \quad (4)$$

In Eq 4 we have the sum of the 5 random values and a scaling coefficient factor is defined by α .

The first five measurement angles are calculated by Eq 5

$$\Theta_i = \Pi * \alpha * r_i, \quad i = 1, \dots, 5. \quad (5)$$

The final angle is calculated by the Eq 6

$$\Theta_6 = \Pi - \sum_{i=1}^5 \Theta_i, \quad \sum_{i=1}^6 \Theta_i = \Pi \quad (6)$$

We have implemented up to the normal floating point representation. To enforce the six angle constraint we have made this restriction explicit. I.e., the function will reject configurations containing anything but 6 validators. Given that the sum of the six measurement angles is constrained to equal Π , the right-hand side of the Θ -protocol parity condition becomes 1 as defined by Eq 7

$$\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \mod 2 = \frac{\Pi}{\Pi} \mod 2 = 1 \quad (7)$$

Therefore, we expect the six-validator experiment to produce an output satisfying the following property:

$$\left(\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \right) \mod 2 = 1 \quad (8)$$

This rather strange-looking design of Eq 8 leads to an extremely simple final consensus condition of just requiring that the XOR of these six measured binary outcomes equals 1.

3.7 Construction of the Six-Qubit GHZ State

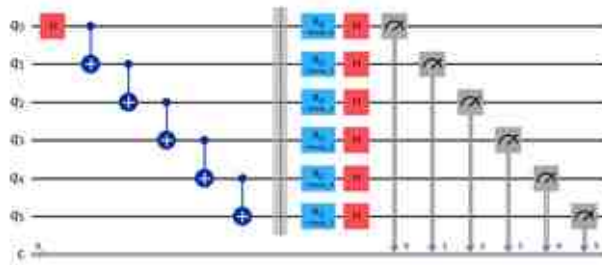


Figure 3: Qiskit Implementation of 6 qubit GHZ circuit

As in the Figure 3, Quantum validation circuit has six qubits and six classical measurement bits $(q_0, q_1 \dots, q_5)$ and $(C_0, c_1 \dots, c_5)$. Initially these are set to $|000000\rangle$.

Then applying a Hadamard gate on the qubit q_0 we get Eq 9,

$$|000000\rangle \rightarrow \frac{1}{\sqrt{2}}|000000\rangle + |100000\rangle \quad (9)$$

Next applying a sequence of linear CNOT entanglement chain of gates yields the desired six qubit GHZ state which gives Eq 10,

$$|GHZ_6\rangle = \frac{1}{\sqrt{2}}(|000000\rangle + |111111\rangle) \quad (10)$$

Instead of implementing six independent circuits for each participant as would be expected from an ordinary GHZ circuit, it's implemented here as a single quantum circuit that creates a commonly shared quantum state between six validators.

3.8 Final Integration into the classical flow

The quantum-validation stage performs measurements in the Θ basis by applying an $R_z(-\Theta_j)$ operation followed by a Hadamard gate H to each validator qubit, yielding a binary measurement result $Y_j \in 0, 1$.

Once computed, the browser application transmits the six measurement angles and the circuit description to the Python-based quantum backend over the network connection. After processing these inputs, the quantum backend returns the resulting six-bit vector as defined in Eq 11, whose result can be mapped to the corresponding validators.

$$\mathbf{Y} = [Y_1, Y_2, Y_3, Y_4, Y_5, Y_6], \quad V_i \leftrightarrow (q_i, \Theta_i, Y_i) \quad (11)$$

These measured bits represent a single quantum-validation event rather than independent approvals. Accordingly, consensus is calculated using the measured parity by Eq 12, where $\oplus$ denotes the exclusive OR (XOR) operation.

$$P_{\text{measured}} = Y_1 \oplus Y_2 \oplus Y_3 \oplus Y_4 \oplus Y_5 \oplus Y_6 \quad (12)$$

The measured parity is compared against the protocol condition and measurement angle constraint which gives $P_{\text{expected}} = 1$ by Eq 13.

$$P_{\text{expected}} = \left(\frac{1}{\Pi} \sum_{j=1}^6 \Theta_j \right) \bmod 2, \quad \sum_{j=1}^6 \Theta_j = \Pi \quad (13)$$

Therefore, for the transaction to transition from the **PENDING** state to the **VALIDATED** state, the measured parity must satisfy the Eq 14.

$$P_{\text{measured}} = P_{\text{expected}} = 1 \quad (14)$$

Finally, after storing the quantum-verification results along with the transaction data itself within our system, the next step involves generating a classical block consisting of the current transaction, its associated previous block hash, and transaction hash, as well as both

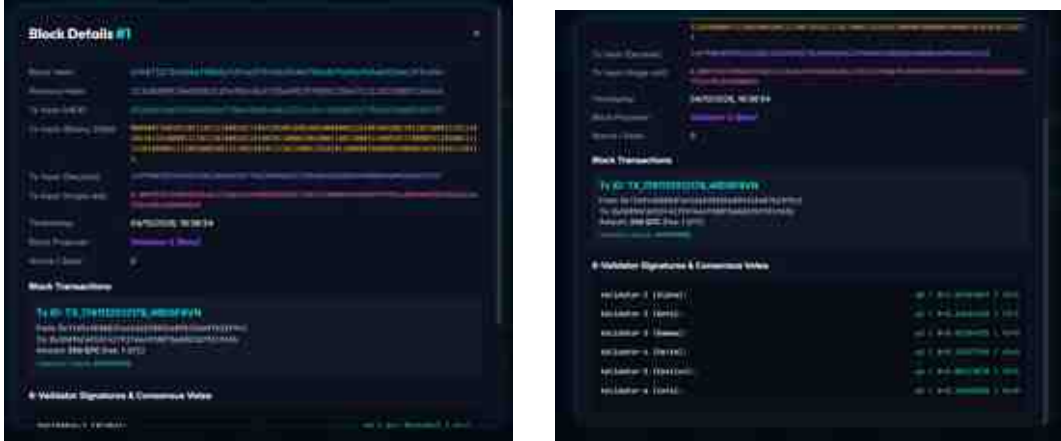
the proposer and the records of verification provided by each validator. This last block can subsequently be secured using SHA-256 hashing. Importantly, though, one should note that the role of proposer comes from deterministic selection among those participating as validators; importantly, this position differs from that randomly chosen verifier who participated during the quantum validation/verification process. With this final step completed, the proposed block becomes part of the shared ledger and is distributed to all participants in the six-validator network. Ledger updates propagate across the distributed system by ensuring that each participant receives an identical copy of the validated block before appending it to their respective local ledger.

Following block propagation, user's balances are updated accordingly, and the contents of the transaction mempool are synchronized with the latest ledger state. To maintain system integrity and ensure correct operation, additional validation checks verify the relationships among the previous block hash, the current block hash, and the transaction hashes. These checks help ensure consistency across the distributed ledger and detect discrepancies between local ledger copies.

4 Results



Figure 4: Shared Ledger



(a) Block information

(b) Block Validator Information

Figure 5: Newly added block details

The prototype successfully demonstrates an end to end transaction validation and block commitment cycle over the 6-node validator network after syncing up with the genesis block. As shown in the figure, when the submitted transaction has been validated with the blockchain protocol, the transaction's SHA-256 hash results into a hash prefix and decimal representation. In this case, we get 0.0097535759 rad as our computed transaction angle. Quantum validation is performed on the Delta node, where a random selection process designates the verifier for the current validation round. Subsequently, six measurement angles are generated such that their combined sum satisfies the required constraint $\sum_{i=1}^6 \Theta_i = \Pi$ radians. The six validator nodes produce the measurement vector as in Eq 15

$$\mathbf{Y} = [1, 1, 0, 0, 1, 0] \quad (15)$$

The measured XOR parity is calculated as

$$P_{\text{measured}} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1 \quad (16)$$

Since the measured parity in Eq16 matches the expected parity, $P_{\text{expected}} = 1$, the Θ -protocol consensus condition is satisfied, allowing the transaction to be accepted for block construction, as illustrated in Figure 5.

Block 1 is subsequently constructed by the Beta node and linked to the genesis block through its previous-block hash reference. Following synchronization across the six validator ledgers, each ledger incorporates the new block, increasing its height from 1 to 2, as illustrated in Figure 4. This was repeated for another 19 times and all the 20 independent simulations trials of the quantum validation achieved a 100% parity since it was in an ideal environment, and the average execution time was $3.25 \pm 0.05S$ per trial.

5 Discussion

The current work is simulator based and does not demonstrate practical security or real-world quantum consensus. While demonstrating successful performance by GHZ based Validation using the above mentioned protocol we found out some constraints which are very crucial from the point of view of actual implementations. First, we see that the Verification condition (collective sum) requires the measurements angles should satisfy certain multiple of π i.e., 0 . An ideal simulator could ensure this constraint. However, real world Quantum Hardware suffers from finite precision control gates and cumulative angle errors leading to possible deviation in the effective sum j thus reducing the reliability of desired parity output. Second, even though the proposed framework provides us with an encoding mechanism for representing the blockchain record as a entangle quantum state, the proposed Θ -protocol only verify the GHZ correlation of the candidate state, instead of verifying the presence of the intended block information in the state. As per the paper's discussion they have described about how their classical block records were encoded into temporal Bell and GHZ States, but the part where you test if your supplied candidate State has the GHZ Correlations required is the Verification Stage here. So basically they showed off their Quantum Validation Mechanism. However, to build up a full-fledged Block Authentication Mechanism, we need to explicitly bind the classical block information we want to authenticate with the Candidate State to be Verified. This becomes really important because someone could simply go onto the Classical Blockchain Network, and tamper the block contents while keeping the underlying Quantum Consensus untouched. And since the Quantum Consensus doesn't care about what happened on the classical side of things, it just happily accepts the tampered block. Now the recent paper regarding Time Entanglements in 2026 [10] manages to overcome this binding challenge, however it runs into serious challenges as the Number Of Blocks Increase, hence working out ways to Combine both methodologies may form an open research problem in our near future scope of Research.

References

- [1] S. Haber and W. S. Stornetta, "How to time-stamp a digital document," *Journal of Cryptology*, vol. 3, pp. 99–111, 1991.
- [2] W. Zhao, "Blockchain technology: Development and prospects," *National Science Review*, vol. 6, pp. 369–373, 2019.
- [3] P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*. IEEE Computer Society Press, 2002.
- [4] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing (STOC '96)*. New York, NY, USA: ACM Press, 1996, pp. 212–219.
- [5] E. O. Kiktenko, N. O. Pozhar, M. N. Anufriev, A. S. Trushechkin, R. R. Yunusov, Y. V. Kurochkin, A. I. Lvovsky, and A. K. Fedorov, "Quantum-secured blockchain," *Quantum Science and Technology*, vol. 3, p. 035004, 2018.

- [6] N. R. Reddy, S. Suryadevara, K. G. R. Reddy, R. Umamaheswari, R. Guttula, and R. Kotoju, “Quantum secured blockchain framework for enhancing post quantum data security,” *Scientific Reports*, vol. 15, p. 31048, 2025.
- [7] K. W. Goh, B. U. I. Khan, A. R. Khan, D. S. Putra, F. R. P. P., and M. A. Bhuyian, “Eliminating single points of trust: A hybrid quantum and post-quantum blockchain with distributed key generation,” *Scientific Reports*, vol. 16, p. 1286, 2026.
- [8] S. Banerjee, A. Mukherjee, and P. K. Panigrahi, “Quantum blockchain using weighted hypergraph states,” *Physical Review Research*, vol. 2, no. 1, 2020.
- [9] D. Rajan and M. Visser, “Quantum blockchain using entanglement in time,” *Quantum Reports*, vol. 1, pp. 3–11, 2019.
- [10] R. Konara, K. D. Zoysa, A. Mahasinghe, A. Sayakkara, and N. Ranasinghe, “Time-entangled quantum blockchain with phase encoding for classical data,” *Quantum Reports*, vol. 8, p. 69, 2026.